

A novel enhanced neural network for anomaly detection in the IoT environment

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ABSTRACT

The Internet of Things (IoT) is changing many fields, including healthcare, transportation, and smart cities. However, the rise of serious cybersecurity issues. This paper introduces a new architecture of Fully connected neural networks (FCNNs) designed for detecting anomalies in IoT settings. The proposed model uses residual connections, multi-head attention mechanisms, SHAP-based feature selection, and RAdam optimisation to improve detection accuracy, robustness, and interpretability. Extensive tests on two benchmark datasets, ToN-IoT and UNSW-NB15, show that the model works well. Specifically, it achieved an accuracy of 99.98% and an F1 score of 99.95% in the ToN-IoT dataset, as well as an accuracy and F1 score of 99.74% in the UNSW-NB15 dataset. These findings suggest that the improved FCNNs architecture can effectively identify anomalies while being lightweight enough for use in real-time IoT systems. The addition of explainability techniques further enhances its usefulness in critical cybersecurity situations by providing precise and interpretable predictions.

1. Introduction

The IoT, a defining characteristic of the IoT is that all elements can be rendered sentient, enabled to have deep digital or virtual representations and interactivity capabilities for communication with devices, and to facilitate communication over the Internet. The integration of various systems with each other creates room for a myriad of creative applications in multiple domains, such as smart homes, healthcare devices, industrial automation, and environmental monitoring [1]. However, the rapid expansion and everyday use of IoT devices also created significant security challenges. The problems stem from IoT networks' heterogeneous and diverse nature of IoT networks connecting various devices with different capabilities with security standards, which allows them to be logical attack vectors [2]. Anomaly detection is a critical component in IoT security [3], it can help improve IoT security by detecting strange behaviour, which may suggest potential threats, an unapproved entry (direct or distributed), a malware contagion, etc. [4]. Typical anomaly detection methods, including rule-based strategies and statistical models, are mostly ineffective for the IoT of the Internet of Things, as there are significant data, complicated network activities, and dynamic IoT environments [5]. The research community has no choice but to turn to more advanced techniques, that is, those based on Machine Learning (ML) and Deep Learning (DL). Recent advances in deep learning have shown an encouraging method for securing IoT devices, as the recent literature proposes, for example, to solve the key synchronisation problem in Low-Power Wide Area Networks (LPWANs) [6]. Deep learning is used at the lowercase level of Deep on the cross-layer level, which helps to augment the authentication link at the upper hand of the deep learning network.

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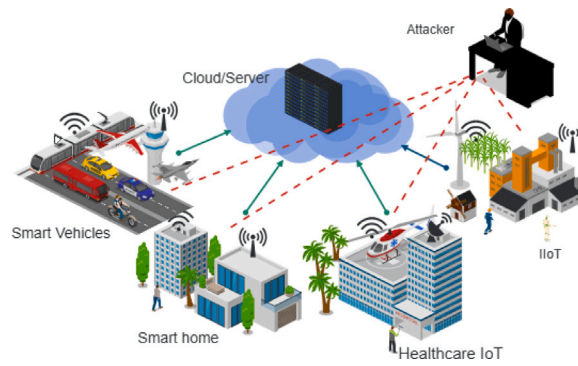


Fig. 1. An overview of IoT environment and security risks.

Fig. 1 illustrates the IoT environment, specifying its components, interactions within the environment, and possible cybersecurity threats. At the highest level is the Cloud/Server, primarily used for collecting, processing, and storing IoT data. The figure depicts four classes of IoT devices — Smart Home (smart locks, enhanced thermostats, etc.), Industrial IoT – machinery/metrology monitoring, Smart Cars, where systems built in a vehicle enable communication between vehicles, and Healthcare IoT, CT or MR machines, etc. An attacker, depicted in the figure, is an external factor that illustrates the weaknesses of the IoT network. Genuine transmission of devices towards the cloud is represented by solid arrows, while dashed red arrows show the possible things that may arise from the attacker. The diagram clarifies the tasks the cloud performs in centre-based processing and especially points out the security threats on the IoT devices and hence evokes great interest for use by academics in discussions, researchers when preparing, and trainers in the materials of presentations dealing with IoT security.

Deep learning, a subset of machine learning, is useful for challenges posed by complex and time-varying IoT data combined with high-dimensionality, non-linearity, and temporal aspects. Provides a dynamic process for improving security through the influence of neural networks [6]. With sufficient amounts of labelled data, deep learning models can learn complex structures and produce accurate predictions without intensive feature engineering [7]. Convolutional Neural Networks (CNNs), among other deep learning techniques, are often used as NSAs on the Internet of Things. In contrast, recurrent neural networks (RNNs) and FCNNs are less noticeable. It allows for the modelling of spatial and temporal relationships and, therefore, is more efficient in detecting less noticeable and complex attacks, which older models could have overlooked [8,9]. Deep learning methods demonstrate advantages in outsmarting and surpassing human capabilities in specific tasks. Some of these challenges can be addressed when using these models for IoT systems. A significant hindrance to this effort is the ability of many IoT devices to support such activities. These devices are often designed with extremely low computational, memory, and battery capacity, which makes it difficult to tolerate the resource-intensive and sophisticated deep learning models on the device [2]. As a result, there is an ever-growing focus on developing smaller and thinner models that work optimally in edge-based devices while maintaining reasonable accuracy levels in anomaly detection [10].

The need to process data in real time is another hurdle that concerns IoT networks. To ensure prompt identification and resolution of threats, anomaly detection paradigms must interpret data from a ‘real-time’ perspective. Consequently, there is a need to build accurate and fast models in which assembling and analysing large volumes of data would not lead to noticeable latency [11]. The use of deep networks in real-time IoT systems also presents some limitations, particularly regarding device scaling and the problem of generated data volume [12]. In addition, deep learning models bring other issues apart from technological ones, such as interpretability. These models have indeed been proven to work with great precision; however, most operate as black-box systems, and thus, it becomes difficult to grasp the principles guiding the predictions made. Insecurity concerns can arise from weak points, as the decision-making process must be understood to create trust and accountability in sensitive applications [13]. Current trends have focused on improving explainable artificial intelligence (XAI), which is a method that explains the reasons behind the operation of deep learning models. This will improve understanding and increase operational efficiency [14].

Considering these difficulties, there is an urgent need for inventive strategies to discover anomalies in the IoT. This paper introduces a customised architecture of FCNNs explicitly designed to detect anomalies in IoT networks. The primary focus of this research is developing and introducing a lightweight, interpretable, fully connected neural network IoT model. Instead of traditional approaches, this model uses Shapley Additive Explanations (SHAP) for explainability, adversarial training for robustness, optimisations for real-time processing, and makes IoT devices easy to use. Our model aims to process complex multidimensional data in tables containing various performance measurements obtained from IoT devices. The main goal is to obtain high precision in detecting abnormal behaviour that may indicate possible security risks while ensuring that the model can be applied to various data sets. To overcome the limitations of IoT devices, we have enhanced the FCNNs model to decrease its computational complexity while maintaining its performance. This optimisation process involves meticulous selection of network design, which includes determining the optimal number of layers and neurones, along with the use of methods such as dropout and batch normalisation to improve generalisation and mitigate overfitting [9]. In addition, the model has been specifically engineered to handle data with high efficiency, allowing the identification of anomalies in real-time in the IoT settings with little delay [10]. Moreover, our model

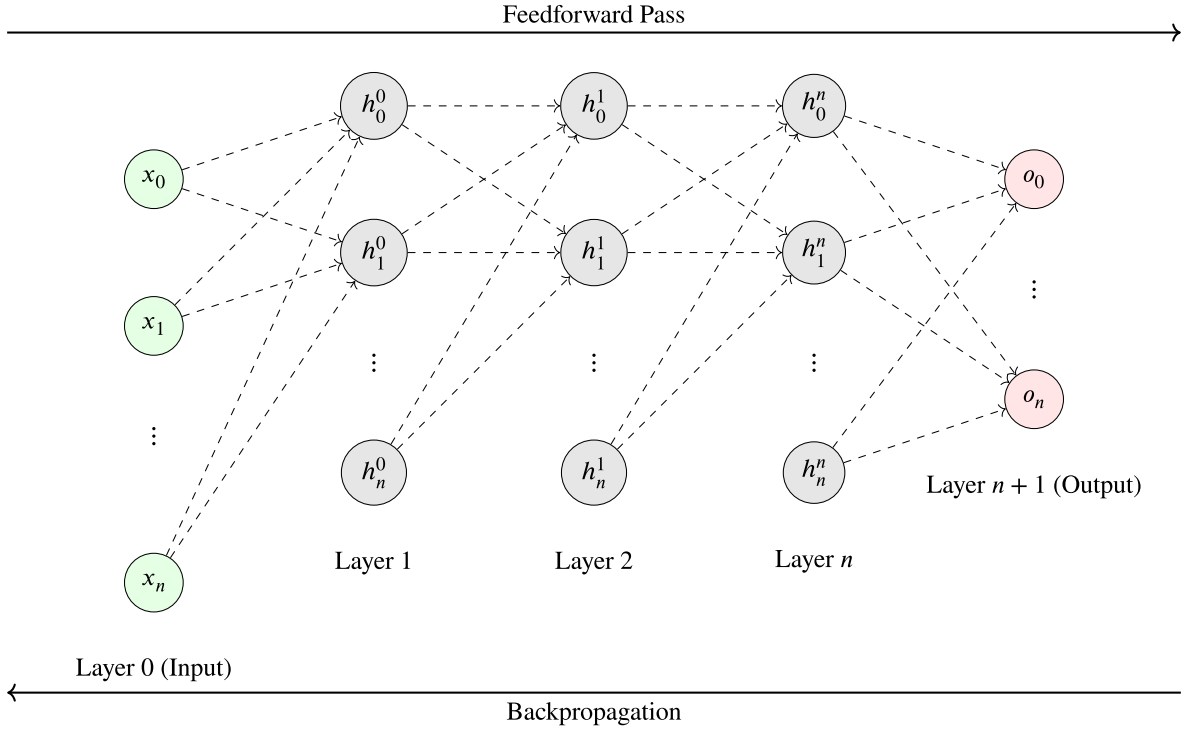


Fig. 2. FCNNs General model architecture.

integrates explainability characteristics that allow users to comprehend and evaluate its predictions. By integrating XAI approaches, we offer valuable information on the specific traits and patterns the model considers crucial in its decision-making process. The openness of the prediction of the model increases confidence and provides significant information to cybersecurity experts to identify and address security issues [13].

Fig. 2 depicts an FCNNs, a general deep neural network used for regression or classification or to perform anomaly or outlier detection. It has an input layer, a number of hidden layers, and an output layer in which each layer is dense, meaning that each node in one layer is connected to every node in the next layer [15]. Layer 0, which is the input layer, receives raw data features (x_0 to x_n), such as sensor values or performance parameters, and sends these to the first hidden layer, layer 1. The hidden layers (h_0^l to h_n^l) serve as intermediate input, using weights and non-linear activation functions, such as ReLU. Let the networks understand the more complex relations in the data. The output layer (Layer $n+1$) produces the final output (o_0 to o_n), typically accompanied by a softmax activation function for classification purposes. Arrows are used to depict the feedforward pass, that is, the direction in which data are transmitted in the network, while arrows are also used to depict backpropagation. An arrow used in this process points in the opposite direction towards the start of the network to update the model's parameters and reduce the error. The presence of ellipses ($\vdots$) within the figure also indicates that the architecture is scalable, allowing it to be extended to include an arbitrary number of nodes and layers, and therefore suitable for multiple tasks. The identification of layers and nodes explicitly helps in theoretical simplification and practical realisation because of the fully connected structure, which suits all predictions of an interrelation between features, and such models find application in varied areas of ML processes.

The proposed FCNNs make a difference in anomaly detection for IoT environments, especially for the typical tabular data encountered in IoT performance metrics. In contrast to CNNs designed for image data that depend on spatial hierarchies [15] or RNNs that deal with sequential data, but have high computational cost and difficulty training deep architectures [16], FCNNs are more appropriate for structured tabular data. Due to the absence of extensive feature engineering, FCNNs are more efficient in capturing heterogeneous, high-dimensional data features for IoT devices [17]. Furthermore, FCNNs are less computationally intensive than CNNs and RNNs, making their deployment in resource-limited IoT devices easier [18]. Their ability to handle tabular data with minimal computation makes FCNNs highly accurate and efficient compared to other deep learning systems, making them the most suitable form of neural networks for detecting IoT anomalies.

The contributions of this study can be summarised as follows.

- The proposed FCNNs model has achieved an impressive 99.95% accuracy rate in the ToN-IoT dataset and 99.74% in the UNSW-NB15 dataset, demonstrating its strength in various IoT settings. Its speed and accuracy make it efficient for IoT devices with limited resources.

- **Demonstration of FCNNs Effectiveness in IoT Security:** This paper shows that fully connected neural networks are efficient when sensing IoT networks for abnormal activities. It contributes to the expanding literature on the use of deep learning in cybersecurity, particularly concerning IoT.
- **Methodological Advancement in IoT Data Preprocessing:** The analysis provides insights into how preprocessing must be performed for IoT datasets, making a valuable addition to the emerging body of knowledge on the methodological aspects of handling and preparing large and complex datasets for machine learning. This may be useful in future research that focuses on the most effective approach to preparing IoT data.
- **Foundation for Real-World Application:** This means that the model performs well and meets the benchmark, thus creating room for deep learning to protect IoT. It is helpful to conduct further research on the application of such models in working environments, which can lead to the development of enhanced practical IoT security models.
- **Encouragement of lightweight, yet effective models:** Thus, research shows that efficient methods for detecting anomalous behaviours of IoT devices and networks can be developed using relatively simple models. This contribution is helpful in academic debates on the trade-off between the level of model detail and the implementation cost, especially in resource-constrained environments such as IoT networks.
- **Potential for Future Research Directions:** The results suggest directions for future studies, such as incorporating advanced neural network structures (e.g., convolutional or recurrent networks) or developing combined methods of FCNNs with other approaches, including unsupervised learning. Moreover, the high performance of the proposed model raises six questions about the feasibility of applying it in other IoT settings and for different datasets, thus increasing its potential usage.

The remaining sections of the paper are organised as follows. Section 2 explores related work on deep learning anomaly detection techniques in IoT networks. Section 3 explains and defines the methodology used for the proposed anomaly detection model. The result and the experimental analysis are explained in detail in Section 4. Section 5 provided a discussion of the proposed model, comparing the model with the previous work, and detailing the implications for IoT cybersecurity. Section 6 offers and defines areas in anomaly detection that need further study. To conclude the paper, a summary of the whole article is provided in Section 7. Fig. 3 shows the organisational structure of the paper.

1.1. Background of the study

The IoT has quickly become an essential technical framework, greatly influencing many industries such as healthcare, transportation, and smart cities. The worldwide anticipated number of IoT devices is expected to exceed 75 billion by 2025, underscoring the widespread incorporation of these technologies into everyday life [19,20]. In addition to fostering innovation and efficiency, this extensive deployment has made IoT systems vulnerable to many cybersecurity risks. The limitations of intrinsic resources in several IoT devices, including restricted computing power, memory, and energy, worsen these weaknesses, making them more susceptible to attacks [21].

IoT networks are distinguished by their dynamic and decentralised structure, producing large volumes of diverse data that conventional cybersecurity methodologies typically find difficult to handle efficiently. The complexity and high-dimensional data environments of IoT networks make traditional security techniques, such as rule-based systems and basic machine learning models, less effective [22]. The generalisation of these approaches across various IoT ecosystems is typically limited, mainly when implemented on devices with limited processing capabilities [23].

The constraints of conventional security approaches underscore the need for well-equipped, sophisticated methodologies to effectively address the unique challenges posed by IoT networks. Due to its ability to represent intricate patterns in extensive datasets, DL has become an auspicious approach to detecting anomalies in IoT settings [24]. Fully Connected Neural Networks FCNNs have demonstrated considerable potential to effectively process structured, tabular data produced by IoT sensors. By capturing complex correlations between characteristics, these models provide a more precise detection of abnormalities that could suggest possible security risks [25]. However, several obstacles remain to the use of FCNNs and other DL models for IoT cybersecurity. A significant obstacle is in achieving a balance between exact prediction accuracy and computing economy, particularly in IoT devices with limited resources [24]. Furthermore, the capacity of these models to adapt to rapidly changing cyber threats and their capacity to expand to large-scale IoT networks are crucial domains that require additional investigation [26].

Transformer models, first designed for natural language processing (NLP), have recently made significant progress and offer a potential solution to these requirements. Transformers have exhibited remarkable proficiency in handling complex and high-dimensional data, and their design allows effective parallel processing [27]. Furthermore, transformer-based models, such as BERT (Bidirectional Encoder Representations from Transformers), have been modified into lighter iterations like DistilBERT and MobileBERT, allowing their use on devices with limited resources without compromising performance [28]. These attributes make transformers suitable for dynamic IoT systems as data and threat landscapes constantly evolve.

This work investigates the use of fully connected neural networks and transformer models to detect anomalies in IoT technology. The research aims to improve the precision, effectiveness and flexibility of cybersecurity measures in IoT networks by leveraging the advanced functionalities of these deep learning methods. The results obtained from this study are anticipated to contribute substantially to the advancement of scalable, resilient, and effective anomaly detection systems that can protect IoT ecosystems from new cyber threats.

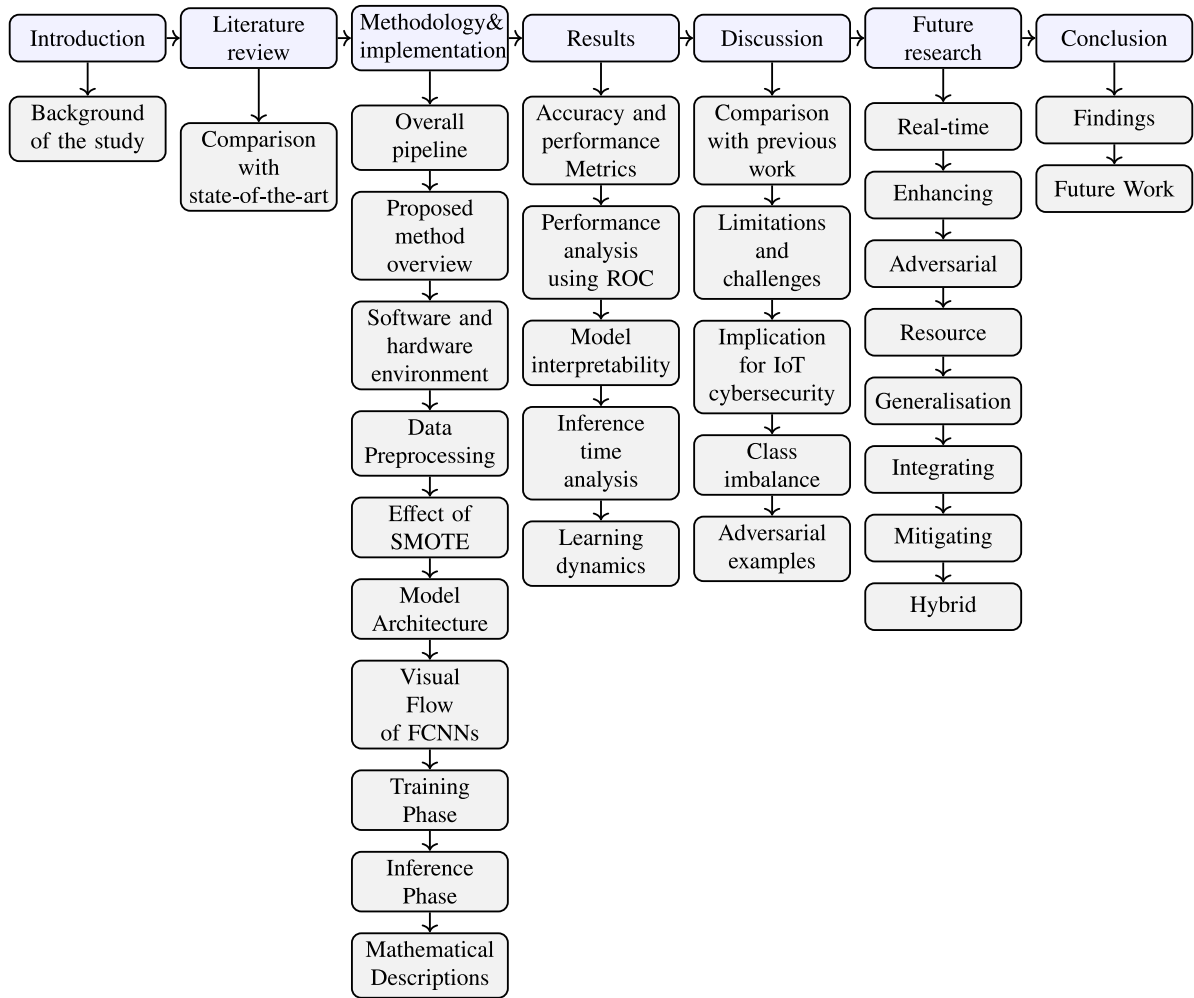


Fig. 3. Paper organisation.

2. Literature review

The exponential growth of IoT devices has dramatically increased the need for solid anomaly detection systems to protect these networks. Alangari [29] adds to this discussion by introducing an autonomous machine learning method specifically designed to identify and detect malicious attacks and abnormal behaviour in devices in the IoT. This strategy is especially beneficial when there is a limited amount of labelled data. Highlights the importance of the algorithm in real-world IoT setups that include a wide range of continuously changing data streams. Altulaihan, Almaiah and Aljughaiman [30] investigated the use of ML techniques in intrusion detection systems (IDS) to counter Denial of Service (DoS) attacks in IoT networks. Their research emphasises the crucial importance of machine learning in strengthening the ability of IoT systems to withstand targeted cyberattacks. This underscores the need for adaptive security solutions.

DL algorithms have become an effective tool for identifying anomalies in the IoT, as demonstrated by several studies that highlight their potential in diverse applications. Benameur et al. [31] provide a DL method incorporated into an intelligent and environmentally friendly irrigation system, highlighting its economical nature and efficacy in identifying abnormalities. This study demonstrates the seamless integration of DL into IoT applications, resulting in improved operational sustainability and security. In their research, Gonaygunta, Nadella, and Pawar [32] provide a versatile DL model designed to improve cybersecurity by improving the identification of anomalies. Their research highlights the potential of deep learning models to handle the intricate and ever-changing characteristics of IoT networks, emphasising their scalability and flexibility. This topic is consistently seen in the existing literature.

The comparative study in [33] thoroughly evaluates several ML techniques for anomaly detection in IoT networks. This analysis offers valuable insights into the advantages and disadvantages of various methodologies. Their work provides an excellent reference for researchers and practitioners choosing the most appropriate techniques to address their difficulties in IoT security. Mishra,

Paliwal and Srivastava [34] utilise deep convolutional generative adversarial networks (DCGAN) to enhance anomaly detection in IoT environments. The use of generative models in their investigation demonstrates the capacity of sophisticated DL approaches to find minor irregularities that may elude conventional detection methods.

Data quality is a crucial part of anomaly detection in IoT networks. Muñoz et al. [35] emphasise this point by introducing an anomaly detection system developed to provide data quality assurance in IoT systems. Their research emphasises the vital importance of high-quality data in the efficiency of anomaly detection systems, specifically to protect the integrity and dependability of IoT operations. Omol, Mburu, and Onyango [36] investigate the use of ML methods for predictive maintenance in smart grids. They show how anomaly detection can improve the efficiency and lifespan of IoT systems in energy management.

Real-time anomaly detection is crucial to ensure the security of IoT systems, especially in networks that rely on microservices and require immediate action to counter potential attacks. Plazas Olaya and Vergara Tejada [37] have created a real-time anomaly detection system that uses ML to enhance the security of microservice-based IoT networks. Their strategy emphasises the importance of scalability and low-latency processing in protecting the security of intricate IoT systems. In this study, Rafique et al. [38] comprehensively analyse the latest developments in ML and DL approaches to detect anomalies in IoT networks. They provide an overview of the current state-of-the-art and highlight the fundamental difficulties that need to be tackled in this field.

According to [39], there is significant progress in improving the use of DL models for IoT traffic categorisation and anomaly detection. They have achieved this by employing deep autoencoders. Their research adds to the continuous advancement of sophisticated models that can effectively manage complex patterns in IoT traffic, thereby improving the precision and dependability of anomaly detection systems. Ullah and Mahmoud [12] explore the process of creating and refining DL models, namely RNNs, to identify anomalies in IoT networks. Their research highlights the importance of choosing suitable neural network structures to balance performance and computational economy, particularly in IoT situations with limited resources.

Wang et al. [7] investigate the use of blockchain technology and deep neural networks to detect anomalies on the Internet of Medical Thing (IoMT). Their novel strategy showcases the ability to combine the decentralised security attributes of the blockchain with the predictive capabilities of DL to safeguard sensitive medical data in IoT systems. Sivapalan, Nundy and Dev [2] introduce Artificial neural network (ANN), a compact neural network specifically developed to detect abnormal patterns in electrocardiogram (ECG) data collected by edge sensors from the IoT. Their model represents the continuous effort to create anomaly detection models that are both efficient and powerful, while still functioning successfully within the limitations of IoT sensors.

2.1. Comparison with state-of-the-art methods

Recent advances in deep learning have led to the development of various models for anomaly recognition within IoT networks. CNNs are widely applied for spatial pattern recognition, which is highly considered for the detection of anomalies in images [40]. However, CNNs are less effective when it comes to processes involving tabular data that are commonly found in IoT systems because they require a great amount of preprocessing and feature engineering to obtain meaningful spatial hierarchies [41]. In the same way, RNNs are useful for sequential data, such as time-series data. However, they have a high computational cost and also suffer from issues such as vanishing gradients, which reduces their effectiveness in real-time IoT systems [42]. However, FCNNs offer greater power and efficiency when handling files with tabular data, a central feature of IoT performance metrics. Without feature engineering, FCNNs can model intricate and nonlinear interdependencies among features, which is helpful in IoT environments where data are primarily high-dimensional and multi-sourced. Furthermore, FCNNs incur lower computational costs compared to CNNs and RNNs, making them more suitable for implementation in resource-poor IoT devices. The suggested neural network model, which is an exact type of FCNNs, takes these factors into account to perform high-precision anomaly detection at minimal economic and computational cost, making it more effective for IoT security tasks than existing techniques. Although CNNs and RNNs have been used for anomaly detection, they are computationally dense and difficult to interpret. Alternatively, the FCNNs model suggested in this study is light-weight and easy to interpret; thus, it is suitable for IoT infrastructures with constrained resources. In Table 1, the proposed model is compared with other existing models, is considered a baseline, and captures better results compared to others in accuracy, computational cost, and interpretability.

3. Methodology and implementation

In this research, we propose an enhanced FCNNs model with residual fusion and multihead attention for anomaly detection in IoT networks. The model is tailored for high-dimensional, tabular data and outperforms all published results on the TON-IoT and UNSW-NB15 datasets. The methodology is divided into three main steps: data preprocessing, model architecture design, and training and testing.

Our experiments used the two datasets to train and evaluate the proposed model as shown in Fig. 5. Telemetry Operating system Network Traffic Dataset (TON)-IoT dataset was collected from the UNSW Canberra, Australia website. The datasets are referred to as 'TON-IoT' since they comprise diverse data sources gathered from IoT and Industrial Internet of Things (IIoT) sensor telemetry datasets, as well as Windows 7 and 10 operating system datasets, and Ubuntu 14 and 18 TLS and network traffic datasets. The data originated from a realistic, expansive network created at the Australian Defence Force Academy's Cyber Range and IoT Labs within the School of Engineering and Information Technology (SEIT) at UNSW Canberra. The dataset is ideal for testing anomaly detection algorithms because it simulates IoT environments [33].

The UNSW-NB15 dataset is one of the most widely used benchmark datasets for evaluating IDS and anomaly detection models in IoT and network security. It was developed by the Cyber Range Lab of the Australian Centre for Cyber Security (ACCS) at the

Table 1

Related works and their contributions.

Ref.	Year	Description	Result	Metric Scores	Implemented Model	Contribution
[43]	2020	An anomaly detection model using deep learning for IoT networks	The result shows an improved detection of IoT anomalies	Precision: 0.91, Recall: 0.89	Deep learning using residual network and Bi-directional Gated Recurrent Unit (GRU)	The main contribution is improving the detection of IoT anomalies using residual network and Bi-directional GRU
[44]	2024	Detection of anomalies in IoT using deep Support vector data description (SVDD) and contractive autoencoders	The result shows that the model can provide a robust detection of anomalies in IoT networks	F1-score: 0.90, Precision: 0.89	Deep SVDD and autoencoder	Robust and efficient anomaly detection in IoT networks
[45]	2021	A graph neural network method for distributed anomaly detection in IoT	The result shows an improved performance in distributed IoT environment	F1-score: 0.93, Accuracy: 0.92	Graph neural network	The article contributed in enhancing the performance of distributed IoT anomaly detection
[4]	2023	Application of cascaded recurrent neural network for anomaly detection in Industrial IoT.	An improved detection of IoT anomalies in industrial applications	Accuracy: 0.92, F1 score: 0.91	Cascaded RNN	The contribution is to enhance the detection of anomalies for industrial IoT environments
[12]	2022	This article describes the flow and flag characteristics used in anomaly detection using a feed-forward neural network	The results demonstrate the model's ability to detect anomalies based on flow and flag characteristics	Accuracy: 99.86%, 99.92%, 99.99%, F1 score: 99.87%, 99.93%, 99.99%	Feed-forward neural network	This article describes flow and flag characteristics used in anomaly detection using a feed-forward neural network
[39]	2024	The article introduces an anomaly detection model to enhance the detection of threats in the IoT environment	The result demonstrated that the employed model can effectively classify and detect anomalies	Accuracy: 95.85%	Convolutional neural network (CNN) and Variational autoencoder (Variational autoencoder (VAE))	This paper improves IoT network anomaly categorisation and detection, notably for network security
[29]	2021	The authors proposed an unsupervised machine-learning algorithm for attack and anomaly detection in IoT sensors.	The result demonstrated effective anomaly detection without labelled data.	Accuracy: 98%	Unsupervised Machine Learning	The paper contributed to unsupervised anomaly detection without labelled data.

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University of New South Wales (UNSW), which aims to model modern network traffic, including both normal and abnormal traffic. The dataset comprises 2,540,044 records with 49 features extracted from network traffic, including flow-based, time-based, and content-based features. These features are designed to capture different behaviour patterns in the network, such as the size of the packet, the type of protocol, and the duration of the connection, which makes the application of the data set in training and testing machine learning models for anomaly detection unquestionable.

In addition to normal traffic, the UNSW-NB15 dataset also captures nine types of intrusions, including DoS attacks, Exploits, Reconnaissance, etc. This variety of intrusion types makes the dataset effective in testing the strength of models for anomaly detection and their capability to address multifaceted attacks. The dataset has been a popular choice among researchers for benchmarking the performance of different machine learning and deep learning algorithms for CNNs and RNNs, as well as ensemble models. It is perhaps the most well-known benchmark dataset for network intrusion detection and is widely used to evaluate anomaly detection systems [48]. Fig. 5 shows a graphical representation of the FCNNs used.

Table 1 (continued).

Ref.	Year	Description	Result	Metric Scores	Implemented Model	Contribution
[8]	2021	The article proposed an IoT botnet anomaly detection using unsupervised deep learning	The result indicated high accuracy in the detection of IoT botnet anomalies	Precision: 99.9% and 91.6%	Unsupervised deep learning	The article's main contribution is the implementation of unsupervised detection of IoT botnet anomalies
[2]	2022	The article proposed a lightweight neural network for detecting ECG anomalies and reducing power consumption in IoT edge sensors	The result demonstrated efficient anomaly detection in edge devices	Accuracy: 97%	Long Short Term Memory Long Short-Term Memory (LSTM) cells and Multilayer perceptron (MLP)	The article contributed to efficient anomaly detection in edge IoT devices
[13]	2024	The article proposed an explainable transformer-based anomaly detection for IoT security	The result portrays an impressive anomaly detection for IoT security	Accuracy: 99%, Recall: 96.28%	Transformer model	The article contributed to implementing an explainable AI model for IoT anomaly detection
[7]	2022	The article proposed anomaly detection in an IoMT-blockchain environment using deep neural networks	The result shows an increased anomaly detection in medical IoT	Accuracy: 87.9% and 84.6%, 89.7% and 85.4%	Deep neural network	The article contributed to proposing an improved anomaly detection in medical IoT environments
[46]	2024	The article proposed a network anomaly detection using graph neural networks	The result demonstrated an improved detection using graph neural networks	Accuracy: 98%	Graph neural network	The paper contributed to anomaly detection with graph neural networks, Graph Attention Network (GAT) and GraphSAGE
[34]	2023	The paper proposed anomaly detection using deep convolutional generative adversarial and bidirectional Long Short-Term memory networks in IoT	The result demonstrated enhanced detection of IoT anomalies with generative models	Accuracy: 99.99%, 99.08%, 99.82%, 99.96%, Recall: 98.90%, 99.45%, 99.99%, 99.07% respectively	Deep convolutional generative adversarial and bidirectional long short-term memory networks	The paper contributed to an enhanced generative model for IoT anomaly detection
[32]	2024	The paper focuses on enhancing cybersecurity using a flexible deep-learning model for enhanced anomaly detection	The result showed enhanced anomaly detection in cybersecurity applications	Accuracy: 99.7%, 99.1%, Precision: 97.1%	Deep learning models	The study proposes a stacked ensemble approach for outlier classification and an enhanced model for anomaly detection in cybersecurity

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3.1. Overall pipeline

The entire pipeline, depicted in Fig. 4, encompasses the workflow as a whole, from raw data collection to model deployment within a real-time IoT security environment. The step begins with data acquisition and preprocessing, in which duplicates are discarded, missing values are imputed, and categorical variables are encoded. Feature importance is first approximated using XGBoost-based SHAP analysis to retain the top 50 most explanatory features. This is followed by training and validation splitting with class balancing using SMOTE to address imbalanced distributions. The preprocessed data are then power transformed and input into the presented FCNN model, which utilises residual blocks and multi-head attention for improved representation learning. Regularisation is achieved through the employment of early stopping and adaptive learning-rate scheduling to prevent overfitting. The model's optimal checkpoint is selected and undergoes a comprehensive evaluation phase, encompassing accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC curves, as well as explainability through SHAP summaries. Finally, the trained

Table 1 (continued).

Ref.	Year	Description	Result	Metric Scores	Implemented Model	Contribution
[9]	2021	The research implemented Deep Neural Network (DNN) for anomaly detection and Mutual Information (MI) for selecting features	The result reveals increased efficiency of anomaly detection with the use of deep learning	Accuracy: 99.010%, Precision: 99.304%, Recall: 98.020%	DNN, CNN-1D, RNN, GRU, and LSTM	The paper contributed to developing a two-stage intrusion detection system for IoT networks and using DNN to detect anomalies in traffic
[47]	2021	The paper focuses on anomaly detection in IoT systems using hyperparameter optimisation and feature reduction	The result demonstrated efficient anomaly detection with deep learning	Accuracy: 99.89%, 99.4%	Feedforward neural network, and autoencoder	The paper's contribution includes integrating diverse IoT traffic instances, analysing and optimising various deep neural networks and identifying a minimal feature set for optimal results
[10]	2023	The paper mainly identifies anomalous behaviour in IoT systems. It protects IoT devices from security attacks due to their weaknesses.	The proposed method is experimentally shown to yield better performance than other models. Moreover, the approach improves identification reliability against unwanted IoT data.	Accuracy: 94.6%	Deep neural network and autoencoder	A better identification of malicious data from compromised IoT devices. Integration of deep learning techniques in achieving high accuracy of the anomaly detection task
[5]	2021	The study identifies anomaly detection in IoT networks as the primary focus of analysis. It effectively addresses binary and multiclass anomalies and uses feature selection to enhance the models.	The proposed model helps detect anomalies in an IoT network. A comparison with previous studies showed that the applied CNN models achieved higher results in anomaly detection.	Accuracy: 99.96%, 99.98%, 99.98%, Recall: 99.95%, 99.96%, 99.95%.	Convolutional neural network 1D, 2D, and 3D.	Anomaly Classification using Convolutional Neural Networks. Applying transfer learning for the binary and multiclass classification problem cases
[11]	2021	The authors present a deep-learning approach to identifying network anomalies, specifically for IoT traffic in the paper	Using a deep neural network architecture indicates enhancements in the attack detection process. However, models require some modifications to detect all attacks accurately.	Accuracy: 98.27%, and 98.29% for training and testing respectively. Recall: 98%	Deep learning model using dense random neural networks	The article contributed to proposing a deep learning neural network approach for Distributed Smart Space Orchestration System (DS2OS) traffic data and an enhanced understanding of IoT security technology and characteristics
This paper	–	proposes a customised FCNNs architecture for anomaly detection in IoT networks.	The result shows that the proposed model can effectively classify and detect anomalies	Accuracy: 99.98%, and 99.74%	Fully connected Neural network	Demonstrated FCNNs effectiveness in IoT security, advanced data processing, and potential future research directions

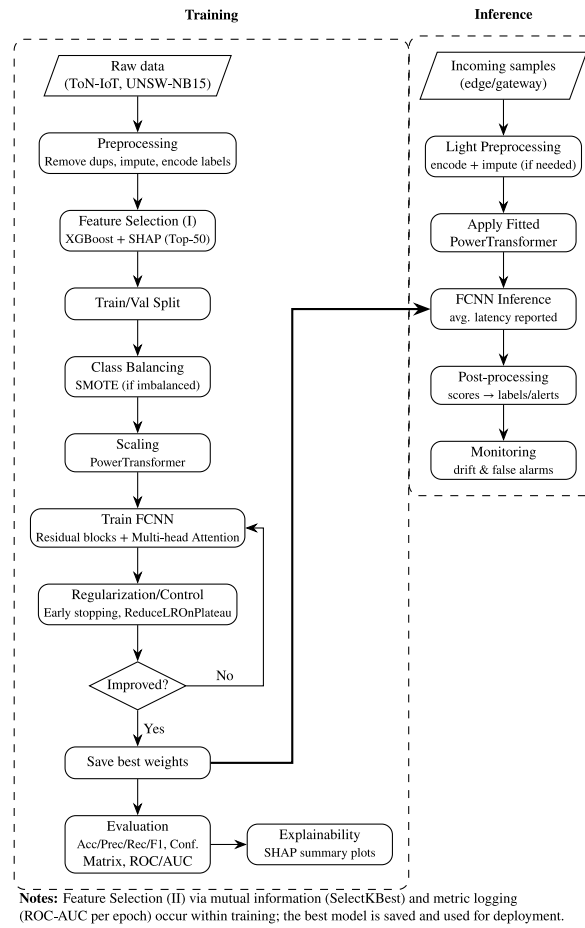


Fig. 4. High-level pipeline of the proposed FCNN-based anomaly detection system. The left pane displays the *training* stages, from data preparation through model selection, regularisation, evaluation, and explainability. The right pane shows the *inference* steps for real-time operation, where the saved best model is applied to incoming samples with minimal preprocessing and monitoring.

model is deployed to the inference pipeline, which inspects incoming IoT traffic with low latency to generate real-time alerts, enabling proactive security monitoring. This top-level figure ensures reproducibility and offers researchers and practitioners a clear, step-by-step description of how the proposed solution can be built and optimised for other IoT scenarios.

3.2. Proposed method overview

The proposed anomaly detector is built on FCNNs, enriched with new functional units developed to improve the depth of learning, generalisability, and explainability of IoT network traffic monitoring, as shown in Fig. 10. These top architecture breakthroughs include:

- **Residual Learning Blocks:** ResNet-based structures motivate residual connections in FCNNs layers to reduce vanishing gradient problems in deep networks. A residual block learns a transform while maintaining the input signal through skip connections, facilitating faster convergence and greater depth in the model architecture. In addition to improving stabilisation in learning, it further enhances feature reuse in layers as a further step towards decreasing overfitting in models.
- **Multi-Head Attention Mechanisms:** Two layers of multi-head self-attention are integrated between residual blocks to learn complex dependencies between features that FCNNs may ignore. These attention layers can be used dynamically to assign weighted values to input features of varying relevance levels for anomaly detection. In this way, the model can differentiate between delicate behavioural patterns of both malignancy and benignancy in high-dimensional IoT data.
- **SHAP-Based Feature Selection:** This framework utilises SHapley Additive Explanations (SHAP) to quantify feature importance in preprocessing. By selecting the 50 features with the highest SHAP values obtained using an XGBoost baseline learning approach, we achieve a reduction in the dimensionality of the input while retaining highly informative features. Not only does it enhance the interpretability of the model, it can also facilitate increased computational tractability without compromising accuracy.

- Class-Imbalance Reduction using SMOTE: Because IoT datasets have an imbalance, minority class examples are over-sampled with the Synthetic Minority Oversampling Technique (SMOTE). It enables learning of distinguishing patterns of under-represented attack types and prevents biased prediction in the majority class directions.
- Hybrid Evaluation Approach: Using a five-fold stratified cross-validation method, along with early stopping and learning rate scheduling, the model's performance is rigorously validated. Performance measures such as accuracy, precision, recall, F1 score, ROC-AUC, as well as confusion matrix visualisation, are considered to have a complete measure.

3.2.1. Innovative contribution

Compared to classical anomaly detection models, which rely solely on hand-engineered features or suffer from the limitations of black-box deep neural networks, this method strikes a balance between accuracy, interpretability, and scalability. That such attention and residual learning in FCNNs is new within this domain assists with improved feature representation, as well as with transparency of learnt behaviour. In addition, incorporating SHAP into this feature selection pipeline introduces a data-driven layer of explainability, thereby enhancing both human trust and regulatory compliance.

3.3. Software and hardware environment

The implementation of the proposed model, together with its evaluation, has been carried out using the following software libraries, together with the hardware infrastructure:

- Programming language:
 - Python 3.10
- Deep learning framework:
 - PyTorch (torch) 2.1.0
- Machine learning and preprocessing libraries:
 - scikit-learn 1.3.0 for preprocessing, feature selection, model evaluation, and
 - imbalanced-learn 0.11.0 for balancing data with
 - xgboost 1.7.6 for gradient boosting and SHAP-based feature importance extraction
 - Shap 0.42.1 for post hoc model interpretability
 - pandas 2.1.0 and numpy 1.24.4 for data handling and numerical computations
 - 3.7.1 and 0.12.2 of matplotlib and seaborn, respectively, for
- Optimiser and activation utilities:
 - RAdam as applied in torch.optim for stable and adaptive learning rate optimisation
 - The softmax function from torch.nn.functional is employed in the normalisation of classification output.
- Development Platform: Google Collaboration (Pro) with GPU-accelerated environment. All scripts run under the Jupyter Notebook interface, with integration of Google Drive for persistent I/O operations.
- Hardware specifications:
 - GPU: NVIDIA Tesla T4 with 16 GB VRAM (Colab cloud back-end)
 - CPU: Dual-core 2.3 GHz virtual processor (host environment)
 - RAM: 25 GB of system memory available at runtime for computations
 - Storage: Google Drive (100 GB quota) is used to access datasets, logs, and model checkpoints.

Each of these trials was run with a PyTorch device selection method that ensures compliance with the CPU and GPU environments. Despite not loading any CUDA library, GPU acceleration was automatically used once available to improve performance in training models and reduce convergence time.

3.4. Data preprocessing

TON-IoT (Telemetry Data from Networks and IoT Devices) is a large and newly proposed dataset intended to benchmark various ML algorithms for classifying different types of cyberattacks in an IoT system. The file Train_Test_Windows_10.csv is a part of the TON-IoT dataset and compiles only the telemetry data gathered from Windows 10 operating systems in an IoT network. The TON-IoT dataset can be considered diverse and multi-input based on network traffic, operating system (OS), and IoT device data, making TON-IoT one of the most realistic and intricate datasets for cybersecurity studies. The TON-IoT dataset is particularly noteworthy because it addresses several key issues in IoT cybersecurity studies, including device heterogeneity, interaction protocols, and the instability of IoT networks. The dataset contains both average and attack data, allowing for the training and testing of models that can classify benign and malicious activities.

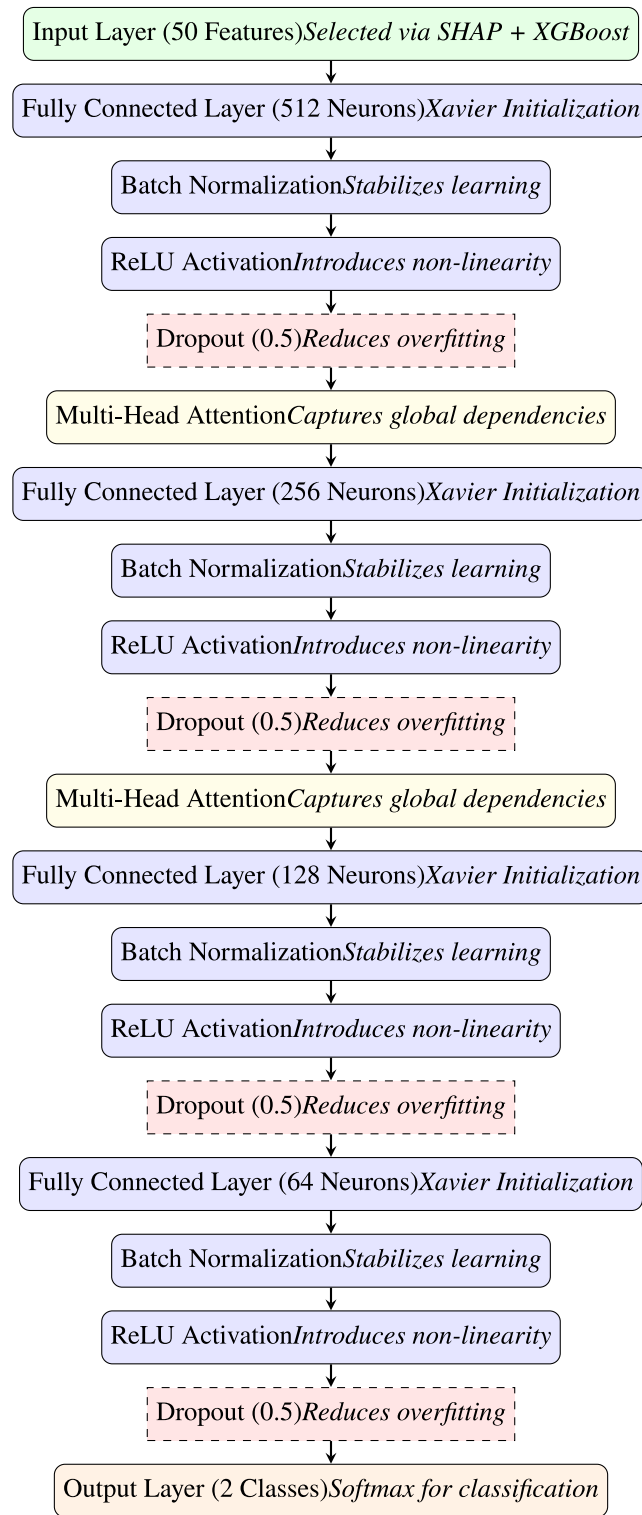


Fig. 5. Visual flow of the FCNNs model architecture.

This study combines the well-known UNSW-NB15 network intrusion dataset with the TON-IoT dataset. The UNSW-NB15 dataset comprises a rich collection of network traffic characteristics, including flow- and packet-based attributes. In addition to normal

traffic, it contains nine categories of modern attacks, including fuzzers, analysis, backdoor, DoS, exploits, generic, reconnaissance, shellcode, and worms. Because this dataset most accurately reflects modern networks and enables the testing of anomaly detection models in various situations, it is frequently used in cybersecurity research. The UNSW-NB15 is one of the most discussed datasets in the literature due to its pose flooding network security threats and attack scenarios, which incorporate pose flood normal traffic scenarios, making detection systems more realistic. These features make the dataset particularly useful for model training to detect a wide variety of cyber threats.

These two datasets offer a different, yet complementary focus: the TON-IoT is limited to device and IoT-specific telemetry, while UNSW-NB15 captures broader network-level anomalies. Together, they enable a comprehensive evaluation of traffic from the device level to the network level in the IoT ecosystem.

The adoption of two data sets ensures that the model is evaluated in a variety of real-world settings, thereby increasing its generalisability and relevance to actual IoT security problems.

The datasets included multiple performance indicators derived from IoT devices, such as central processing unit (CPU) idle time, disk write rate, and other characteristics of the system.

Columns with zero values: The columns limiting the variation in a sample to zero were removed from the dataset. Without any variability, these columns do not provide any additional information that could aid in learning in separate classes.

Handling Missing Values: Assignment of non-numerical symbols a value of not a number.

- Where data were out of the context of numeric interpretation, non-numeric values were exchanged with NaN.
- Utilisation of Column Medians to Provide Missing Numeric Values:
- For columns that were numeric and had missing values, the median of the columns of the cells was used. The medium is used instead of the average, as it is a more robust measure and less sensitive to extreme values, which can adversely affect the model.

Deletion of columns without numerical values: The dataset was stripped of non-numeric columns. The model works on numerical data, and all non-numeric columns require further processing to be usable (e.g., one-hot encoding). A more straightforward option would be to delete the columns irrelevant or redundant to classification.

Hybrid feature selection: The 50 best features were selected using a hybrid method that integrates SHAP and XGBoost. SHAP values grant interpretable feature importance, and XGBoost performs powerful ensemble methods for feature selection. The technique reduces the dimensionality of the data set, improving the computation time and minimising the potential for overfitting.

Data normalisation: To normalise the data, PowerTransformer was implemented, which allows all features to be on the same scale. This is necessary to improve convergence with gradient optimisation-based algorithms and allows the model to treat all features with equal importance.

3.5. Effect of SMOTE on class distribution

To address the imbalance in the datasets, Figs. 6 to 9 demonstrate how SMOTE helps mitigate the class imbalance in both datasets examined in this study. As can be observed in Fig. 6, the ToN-IoT dataset is initially mildly imbalanced; however, after applying SMOTE (Fig. 7), there is an approximately equal representation of standard and attack samples. This balance is particularly crucial for the UNSW-NB15 dataset, where the pre-SMOTE distribution (Fig. 8) is heavily imbalanced in favour of attack classes, potentially causing the classifier to be biased towards the predictions of the majority class. With SMOTE implemented, the distribution in Fig. 9 is symmetric, and the FCNN model is exposed to balanced training data. This step is instrumental in decreasing overfitting by the majority class and increasing recall for minority attack classes, which is one of the main reasons for the high F1-scores and precision in our analysis. With these visual comparisons included, we improve the interpretability of our preprocessing pipeline and present a key methodological contribution to preprocessing data for successful anomaly detection in IoT networks.

3.6. Model architecture

The FCNN model is designed to consider residual connections and multihead attention mechanisms, which enable the model to fully utilise the complexity of IoT data. Everything, from computational costs to architectural designs, is taken into account when dealing with multidimensional, tabular data. The Fig. 11 presented gives a step-by-step structural diagram of an FCNNs map suited for classification and clearly defined in this IoT anomaly detection. This figure illustrates the network's layers: the input layer, fully connected hidden layers, ReLU activation functions, and the output layer.

- Input layer: The layer receives the input consisting of feature vectors that have been normalised with respect to the performance measures of the IoT devices. The layer contains 50 neurones, which represent the top 50 features selected during the preprocessing stage.
- Residual blocks: The input layer takes in the normalised feature vectors pertaining to the performance indices of the IoT devices. The input layer contains fifty neurones that represent the top fifty features selected during preprocessing. So, all features are automatically updated for every preprocessed dataset.
 - First residual block: 502 Recurrent neural network layer RNNs with batch normalisation, dropout (rate = 0.5), and ReLU activation.

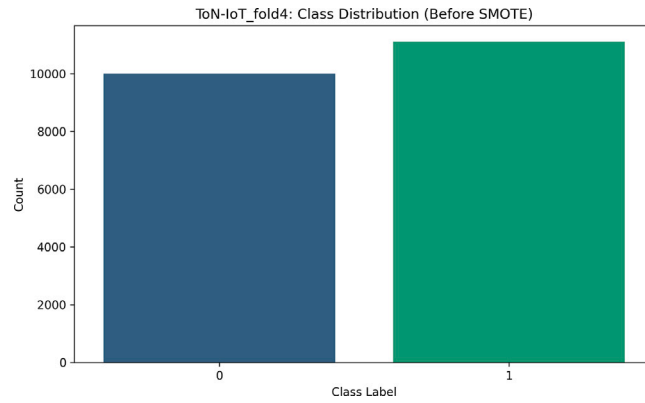


Fig. 6. Class distribution of ToN-IoT dataset before SMOTE, showing a mild imbalance between normal (class 0) and attack (class 1) samples.

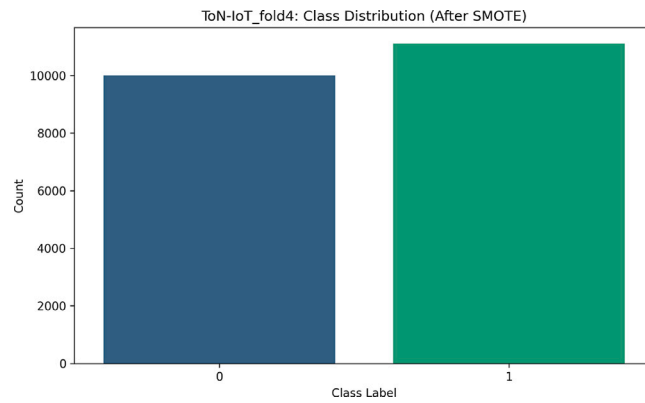


Fig. 7. Class distribution of ToN-IoT dataset after SMOTE, where minority and majority classes are brought closer to parity.

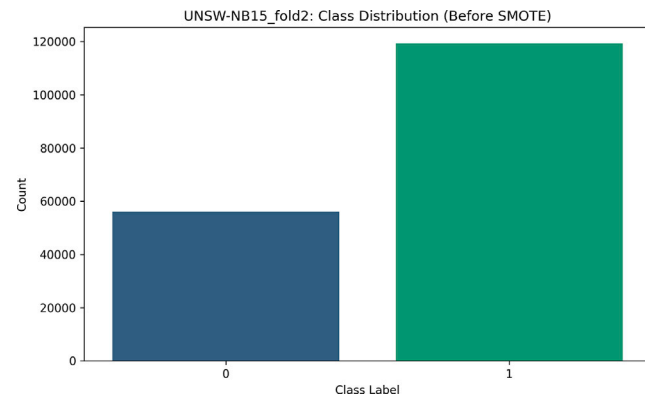


Fig. 8. Class distribution of UNSW-NB15 dataset before SMOTE, revealing a significant skew with normal traffic (class 0) being substantially fewer than attack traffic (class 1).

- Second residual block: 256 Denote a fully connected layer, followed by batch normalisation, ReLU activation, and dropout (rate = 0.5).
- Third residual block: 128 Fully connected layer, followed by batch normalisation, ReLU activation, and dropout (rate = 0.5).
- Fourth residual block: 64 neurones with batch normalisation, ReLU activation and dropout (rate = 0.5) and a fully connected layer.

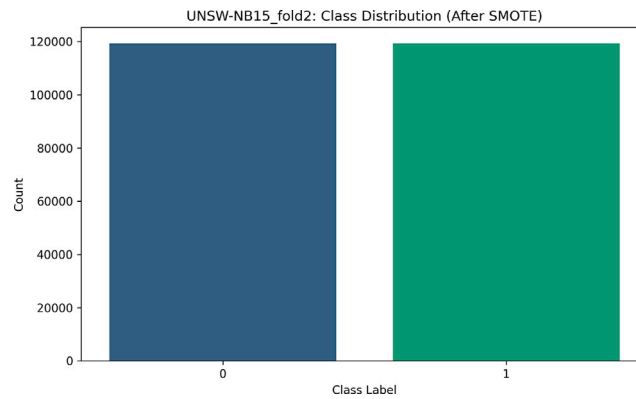


Fig. 9. Class distribution of UNSW-NB15 dataset after SMOTE, demonstrating that SMOTE effectively balanced the class counts, resulting in nearly equal representation.

- **Multi-Head Attention:** Multi-head attention blocks are introduced in between the residual blocks to capture the global interactions within the data. Such layers enable the model to refine its focus on the most significant aspects of subtle anomalies, which can be easily overlooked, for detection.
- **Output Layer:** The last layer, which is the output, is a completely integrated layer that is also activated with softmax. It gives the probabilities of the defined classes being normal or during an anomaly. The output of the softmax function is always between zero and one, and its summation will always be equal to one, so it is perfect for two-class classification problems.

3.7. Visual flow of the FCNNs model architecture

Fig. 5 displays a schematic representation of the FCNNs model, which takes one from the input layer and traverses through the hidden layers to the output. Residual connection and multihead attention mechanisms are implemented to enhance the model's ability to capture complex patterns in IoT data.

- **Feedforward process:** The arrows in the figure denoting the interrelations between layers depict the feedforward process, which describes the movement of data throughout the network. It starts from the input layer where the feature vectors pertaining to the performance metrics of IoT devices are normalised are given to the model. The data is then transformed by passing through several residual blocks. These blocks comprise a fully connected layer, batch normalisation, ReLU activation, and dropout. The model features residual connections that facilitate the generation of deeper representations, thereby alleviating the problem of vanishing gradients. Multi-head attention layers are added in between the residual blocks to grab the global dependencies in the data. These layers enable the model to focus more on the key defining features of an anomaly, thereby mitigating its overall ability to identify subtle nuances within the anomalies. The last layer is a fully connected layer with a softmax activation function that outputs a probability distribution.
- **Backpropagation processes:** The figure also illustrates the backpropagation process, which determines how errors are computed based on the differences between the model predictions and the actual results. These errors are fed backward into the network, and the weights are modified with the use of the RAdam optimiser. The process continues in a loop until the model converges to a defined consensus, while ensuring that the weights are pushed towards the point of the lowest function value.
- **Layer descriptions:** Each layer in the model is illustrated with rectangles, with additional notes explaining the purpose of each layer. For example:
 - **Fully Connected Layer (512 Neurones):** The hidden first layer is fully connected to 512 neurones and was initialised with Xavier.
 - **Batch Normalisation:** Helps to stabilise the learning process by normalising the input of each layer.
 - **ReLU Activation:** Provides a non-linear process to the model so that it learns and captures complex structures.
 - **Dropout (0.5):** Aids in reducing overfitting by randomly switching off neurones during training.
 - **Multi-Head Attention:** Supports the model in focusing more on the most crucial feature by identifying and capturing the global dependencies in the data.
- **Softmax activation and output:** The output layer block underlines the presence of the softmax activation function, which is commonly used in cases where the classification problem involves multiple classes. This layer produces a distribution over two classes (Normal and Anomaly).

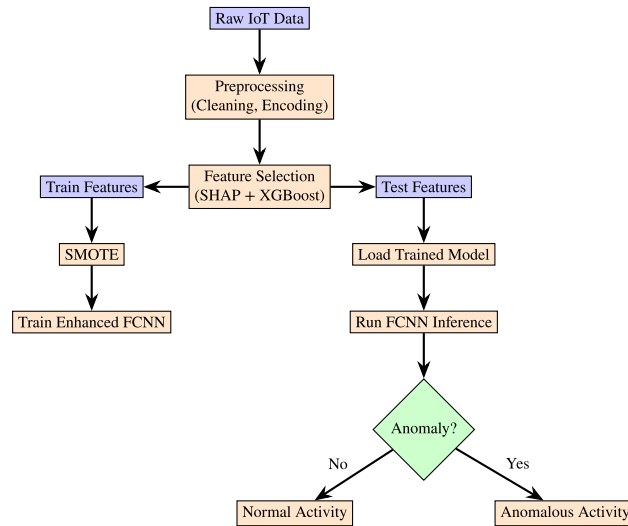


Fig. 10. System-level block diagram of the proposed FCNN-based IoT anomaly detection pipeline. It consists of both training and inference phases, utilising SHAP-based feature selection, class rebalancing with SMOTE, and explainable anomaly classification.

- Interpretation of output: Neural networks feature a single final layer that can generate output for either one or more classes. Whichever class the model assigns its probabilities to needs to be positively declared; therefore, the class with the highest probability is selected as the predicted class. In this case, the model identifies an anomaly when the probability of an anomalous class exceeds that of the regular class. Using this approach enables the model to issue confidence in its predictions, which is very useful in cybersecurity situations due to the need to manage false positives and false negatives effectively.

3.8. Training phase: Learning normal and anomalous pattern

During training sessions, we used an optimised learning technique called RAdam, also known as Rectified Adam, as shown in Table 2. This technique is more sophisticated than the traditional Adam Optimiser, as it works to fix the variance that arises when setting an adaptive learning rate. This accounts for the issues that result from using the traditional Adam. With this reward technique, the stability of adversarially learnt systems is improved, making it ideal for training deep neural networks. Cross-entropy loss was used as the loss function as it is best suited for binary classification, which this particular task requires. Additionally, a more sophisticated scheduling method was employed to enhance training efficiency. The learning rate is halted whenever the accuracy does stop improving, thus smoothing the gradient and avoiding over-learning of the system.

The system was trained in 100 epochs with a batch size of 32. The overfitting was monitored through training and the validation losses, in which the weights of the classes within the loss function were also incorporated into the model. These modifications mitigate class imbalance by targeting the least represented classes. Steady convergence during the training process is a positive indicator, as is the decrease in training and validation losses. These results indicate that effective learning occurred without an overthreshold. Furthermore, applying dropout (kept at 0.5) and batch normalisation improved the model's generalisation performance in the new data.

The architecture features all interconnected layers, comprising an input layer, two hidden layers, and an output layer. Based on prior research [5,49], a dropout rate of 0.5 was selected due to its ability to reduce overfitting without significantly degrading model accuracy. The layer sizes (512, 256, 128, 64) were chosen through ablation studies, where we explored various other architectures and their corresponding performances in terms of the computational resources required. The final architecture was selected to achieve maximum accuracy with optimal resource usage. To provide some level of explainability, we added SHAP (SHapley Additive Explanations) [13,14] so that some reasoning could be provided for the model predictions. The corresponding SHAP values provide the rationale for flagging features as anomalous, highlighting each feature's contribution to the prediction. In dealing with the issue of class imbalance, we used the SMOTE approach, or Synthetic Minority Oversampling Technique [50], which involves the creation of synthetic samples for the minority class through interpolation. This better balances the majority and minority classes, increasing the likelihood of accurately identifying rare anomalies. Fig. 10 shows a system-level block diagram of the proposed model, which further provides a visual description of the model.

During the training phase, the system provides labelled data on normal and anomalous behaviours for the model. This training must enable the model to distinguish between these two classes based on their characteristics (e.g. CPU usage, disk activity, etc.).

- **Feature Representation:** The model begins by accepting the input data as feature vectors, each representing a frame of the IoT device state. Such vectors may include numerical values that measure different attributes of the device's performance. All input features are standardised using the PowerTransformer to have the same impact for every related attribute.
- **Forward propagation:** Subsequently, the input vector passes through other layers in the neural network. Both layers consist of a linear transform using matrix multiplication, the addition of a bias term, and nonlinearity using rectified linear units (ReLU). The network also learns how to combine the features to arrive at a configuration with the highest potential to discriminate between normal and anomalous classes.
- **Loss calculation:** The predictability of the model is determined by cross-compiling the result with the original labels of the dataset using the CrossEntropyLoss loss function. This loss function finds the mean of the absolute difference between the predicted probability and the actual class. The idea here is to minimise this loss, suggesting that the model's predictions on the correct labels should be closer to the ground truth with each cycle of the training data.
- **Backpropagation and weight update:** During backpropagation, the loss gradients concerning the network weights are calculated using the chain rule. These gradients are used to update the weights through the RAdam optimiser, where the loss is reduced in combination with adapting the learning rates, making it the best of both worlds of stochastic gradient descent (SGD). The ReduceLROnPlateau scheduler increases the convergence efficiency by reducing the learning rate when the validation accuracy plateaus. This will be repeated until the model relates certain input features to normal or abnormal activity with the desired accuracy.

3.9. Inference phase: Detecting anomalies

During inference, the model operates on unseen IoT feature vectors by feeding forward with learnt weights and biases. Logits from the output layer are transformed into probabilities with softmax. The input is finally labelled as normal or anomalous, depending on the highest probability. This lightweight inference architecture can be deployed in real time in resource-constrained IoT settings.

- **Input data:** The model gets a new feature vector corresponding to an IoT device's current state. This would be real-time data or data from the most recent operations to assist the planner.
- **Forward propagation:** Mathematical operations such as multiply and add are applied to the input vector through the neural network layers with learnt weights. The network performs the same transformations learnt in the training process and produces an output.
- **Logits and Class Probabilities:** The vectors that emerge from the network are the logits. These raw scores indicate how strongly the model associates the input with the current class, whether it is normal or anomalous. More often than not, these logits are passed through a softmax function to result in probabilities. The probability distribution indicates the percentage chance that the given input belongs to any class.
- **Classification Decision:** The model can predict the class having the highest probability among all classes in the dataset. If the probability for the class categorised as 'anomalous' is more significant than for the class marked as 'normal', the model assigns the input as an anomaly.

The model's performance was evaluated using several key metrics.

- **Accuracy:** The first and primary measure was obtained: $\text{accuracy} = (\text{number of instances rightly classified}) / (\text{number of cases in the test set})$.
- **Confusion matrix:** A confusion matrix was created, which gives ideas about the nature of mistakes made by defining the proper categories of the model as true positive, true negative, false positive, and false negative variables.
- **Classification Report:** Accuracy, precision, recall, and the F1 score were used as metrics to evaluate the performance of the model.

3.10. Mathematical description of FCNNs

A fully connected neural network FCNNs comprises multiple layers of neurones, where each neurone in one layer is fully connected to all neurons in the subsequent layer. The following presents the mathematical formulation of the model.

3.10.1. Feedforward process

The output of the FCNNs is computed layer-wise, starting from the input layer. Let:

- L denote the total number of layers (input, hidden, output),
- $\mathbf{h}^{(l)}$ represent the activation vector at layer l ,
- $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ denote the weight matrix and bias vector at layer l ,
- $g(\cdot)$ be the non-linear activation function (e.g., ReLU, Sigmoid).

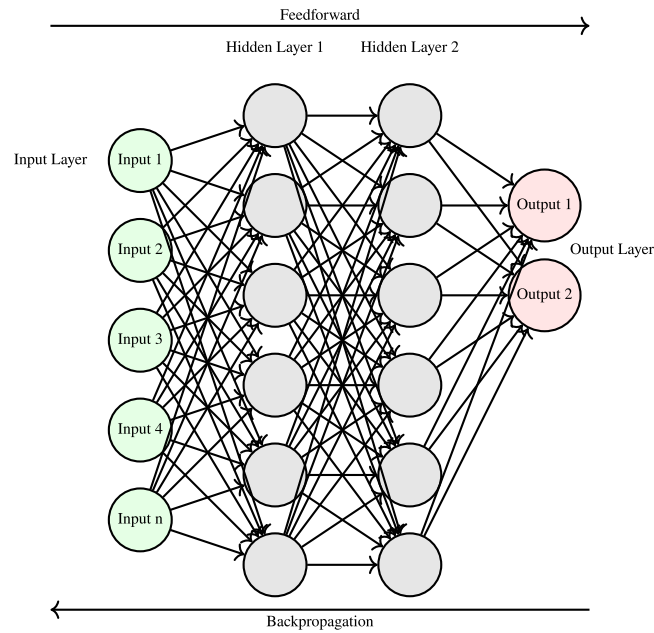


Fig. 11. Structural representation of the proposed FCNNs. It is a densely connected network where all the nodes from one layer are connected to all nodes in the next layer. The architecture uses residual connections and batch normalisation. Feedforward and backpropagation direction paths are shown for the training and inference stages.

Table 2

Parameters used for analysis on FCNNs model.

Parameter Category	Parameter	Value/Description
Data preprocessing	Feature selection method	SHAP with XGBoost; Mutual Information (SelectKBest)
–	Label Encoding	Sklearn LabelEncoder
–	Imputation Strategy	Median Imputation (SimpleImputer)
–	Scaling Method	Power Transformer
–	Data Balancing Technique	SMOTE (Synthetic Minority Over-sampling Technique)
–	Number of Selected Features	Top 50 features (based on SHAP)
Model Architecture	Base Model	Fully Connected Neural Network (FCNN) with Residual Blocks
–	Attention Mechanism	Multi-Head Attention (2 layers)
–	Number of Attention Heads	4 per attention layer
–	Residual Layers	4 layers with [512, 256, 128, 64] neurons respectively
–	Activation Function	ReLU
–	Dropout Rate	0.5 per residual block
–	Final Output Layer	Softmax layer for multi-class classification
Training Settings	Optimiser	RAadam (Rectified Adam)
–	Learning Rate	0.001
–	Batch Size	Full batch (all samples per epoch)
–	Number of Epochs	100 (with early stopping based on validation performance)
–	Early Stopping Patience	10 epochs
–	Learning Rate Scheduler	ReduceLROnPlateau, factor=0.5, patience=5
–	Loss Function	Weighted CrossEntropyLoss (based on class imbalance)
Evaluation Strategy	Cross-Validation	5-Fold Stratified K-Fold
–	Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC-AUC
–	Best Fold Selection Criterion	Highest validation accuracy
Visualisation and Explainability	Feature Importance Plot	SHAP Summary Plot
–	Model Evaluation Visualisation	Confusion Matrix, ROC Curve, Accuracy and Loss Curves

The forward pass is computed as:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}, \quad l = 1, \dots, L-1 \quad (1)$$

$$\mathbf{h}^{(l)} = g(\mathbf{z}^{(l)}), \quad l = 1, \dots, L-1 \quad (2)$$

In the output layer L , a softmax activation is applied:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z}^{(L)}) = \frac{\exp(\mathbf{z}^{(L)})}{\sum_j \exp(z_j^{(L)})} \quad (3)$$

Eq. (1) computes the weighted sum of the inputs in layer l using the weight matrix $\mathbf{W}^{(l)}$ and the bias vector $\mathbf{b}^{(l)}$. The resulting linear combination $\mathbf{z}^{(l)}$ is then passed through a nonlinear activation function $g(\cdot)$ (typically ReLU or LeakyReLU) as shown in Eq. (2), which produces the activation vector $\mathbf{h}^{(l)}$.

Eq. (3) applies the softmax function to the output layer (L), converting raw logits $\mathbf{z}^{(L)}$ into a probability distribution $\hat{\mathbf{y}}$ over C classes. This is crucial for multiclass classification tasks, such as distinguishing between various types of IoT attacks.

3.10.2. Residual connections

Residual connections improve gradient flow by adding identity mappings. The output of a residual block at layer l is:

$$\mathbf{h}^{(l)} = g(\mathbf{z}^{(l)}) + \mathbf{h}^{(l-1)} \quad (4)$$

where $\mathbf{z}^{(l)}$ is computed as in the standard feedforward step [51].

Eq. (4) implements residual learning, where the input of the previous layer $\mathbf{h}^{(l-1)}$ is added to the non-linear transformation of the output of the current layer. This skip connection mitigates vanishing gradient issues and allows the model to train deeper architectures by enabling identity mappings.

3.10.3. Multi-head attention

Multi-head attention captures long-range dependencies. For an input $\mathbf{h}^{(l)}$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (5)$$

with:

- $\mathbf{Q} = \mathbf{W}_Q \mathbf{h}^{(l)}$, $\mathbf{K} = \mathbf{W}_K \mathbf{h}^{(l)}$, $\mathbf{V} = \mathbf{W}_V \mathbf{h}^{(l)}$, and
- d_k as the dimensionality of the key vectors [27].

Eq. (5) defines the scaled attention of the dot product used in the multihead attention mechanism. The query matrices ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) are linearly projected versions of the input $\mathbf{h}^{(l)}$. The attention score reflects the relevance of each token to others, normalised via softmax. This mechanism enables the network to capture temporal dependencies and interactions among features, improving anomaly pattern detection.

3.10.4. Loss function

Cross-entropy is used for classification:

$$\mathcal{L}(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (6)$$

where C is the number of classes, $\mathbf{y}$ is the hot label, and $\hat{\mathbf{y}}$ is the predicted probability [15].

Eq. (6) calculates the cross-entropy loss, which quantifies the difference between the predicted probability distribution $\hat{\mathbf{y}}$ and the ground truth label $\mathbf{y}$. The use of this loss function is appropriate for classification tasks and ensures that incorrect predictions are heavily penalised.

3.10.5. Backpropagation and parameter updates

Gradients are computed by applying the chain rule backward through the network [15]:

Error Signal:

$$\delta^{(L)} = \hat{\mathbf{y}} - \mathbf{y} \quad (7)$$

$$\delta^{(l)} = (\mathbf{W}^{(l+1)})^T \delta^{(l+1)} \odot g'(\mathbf{z}^{(l)}), \quad l = L-1, \dots, 1 \quad (8)$$

Gradients:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} = \delta^{(l)} (\mathbf{h}^{(l-1)})^T \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} = \delta^{(l)} \quad (10)$$

Weight Updates:

$$\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}}, \quad \mathbf{b}^{(l)} \leftarrow \mathbf{b}^{(l)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} \quad (11)$$

Table 3
FCNNs model result.

Datasets	Accuracy%	Precision%	Recall%	F1 Score%
TON-IoT	99.98	99.91	1.00	99.95
UNSW-N15	99.74	99.71	99.77	99.74

Eq. (7) initialises the error signal in the output layer as the difference between predicted probabilities and true labels.

Eq. (8) propagates the error backwards through the network using the derivative $g'(z^{(l)})$ of the activation function. The term $\odot$ represents the Hadamard (element-wise) product.

Eqs. (9) and (10) compute the loss gradients with respect to weights and biases. These are used to update the model parameters during training.

Eq. (11) updates the parameters $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ using the calculated gradients and the learning rate η , following the gradient descent rule.

3.10.6. Final prediction

The final class prediction is selected as [15]:

$$\hat{y} = \arg \max_i \hat{y}_i \quad (12)$$

Eq. (12) selects the class label with the highest predicted probability, representing the final decision of the FCNNs model.

3.10.7. Research context

Collectively, these equations describe the forward pass, attention enhancement, loss computation, and back propagation for the FCNNs model. Together, these enable effective learning and accurate classification of complex high-dimensional IoT traffic data into benign and malicious labels.

4. Experimental results

The evaluation demonstrated that the proposed model was superior in terms of the measures chosen, including high precision. This confirms the high efficiency of the proposed model in identifying anomalous behaviour patterns in the traffic of IoT networks. The latter F1 score also explains the model's ability to achieve a good midpoint between precision and recall, making it useful for anomaly detection in cybersecurity. The improved recall of minority attack classes is attributed to the well-balanced training data guaranteed by SMOTE (see Figs. 6 to 9). SHAP analysis revealed that features related to network flow, including packet number, byte size, and connection time, were the most influential factors in model predictions. These are good indicators of anomalous behaviour, as most IoT attacks, such as DDoS, port scanning, or data exfiltration, manifest as abnormal traffic spikes or unusual session durations. By prioritising these traffic-level features, our FCNNs not only accurately detect attacks but also align with operationally meaningful signals that security professionals monitor, which increases the interpretability and reliability of the model in actual IoT deployments.

Based on the findings depicted in the objective test of the FCNNs model, the value obtained in anomaly detection of the IoT network plays an indicative role in the general reliability and efficiency of the model. The evaluation parameters chosen, accuracy, precision, recall, F1 score, and confusion matrix, all testify that the proposed model is highly efficient. Let:

- TP : True Positives
- FP : False Positives
- FN : False Negatives
- TN : True Negatives

4.1. Accuracy and performance metrics

The model achieves a total accuracy of 99.98% and 99.74%, as shown in Table 3, by classifying the given data. This high level of precision shows that the proposed model is relatively successful in detecting normal and abnormal behaviours originating from the IoT network data. This implies that the model successfully classifies nearly all instances of the two classes with accuracy, without any confusion between them.

Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

As can be seen, normal data points (class 0) and anomalous data points (class 1) achieve accuracy scores of 99.98% and 99.74%, respectively. The result means that nearly all the data points classified as anomalies (or normals) by the model were indeed anomalies (or normals).

Precision measures the ratio of the number of accurately predicted positive instances to the total number of instances predicted as positive. It is a crucial metric in anomaly detection applications where false positives can incur significant costs. A model with a

Table 4
Confusion matrix for the ToN-IoT dataset (binary classification).

-	Predicted normal	Predicted attack
Actual normal	1998	2
Actual attack	0	2221

Table 5
Confusion matrix for the UNSW-N15 dataset (binary classification).

-	Predicted normal	Predicted attack
Actual normal	23 798	70
Actual attack	56	23 812

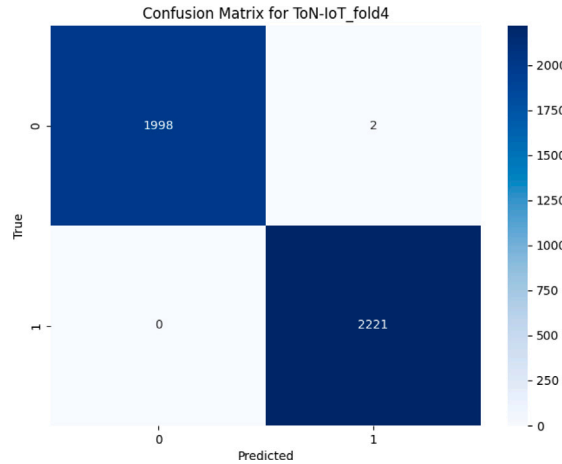


Fig. 12. Confusion matrix of the FCNNs model on the ToN-IoT dataset: The matrix portrays the classification results for the ToN-IoT dataset using the proposed FCNNs model. The model achieved 1,998 true positives, 2,221 true negatives, two false positives and zero false negatives, achieving high precision and recall.

high precision score is shown to be conservative for positive predictions, never mistakenly labelling benign samples as attacks. In this research work, the designed FCNN model achieved precision values of 99.91% and 99.71% for the ToN-IoT and UNSW-NB15 datasets, respectively, demonstrating its excellent ability to identify malicious activity while generating very few false positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity) reflects the ability to find all relevant instances:

$$\text{Recall} = \frac{TP}{TP + FN}$$

The recall scores were 100% and 99.77%. These results, depicted in Table 3, showed that the developed model correctly identified almost all actual anomalies and the normal data points in both classes. This is particularly important in IoT security, as failure to detect an anomaly, considered a false negative, means that security breaches are not detected, which can be disastrous.

F1-Score is the harmonic mean of precision and recall:

$$\text{F1-score} = \frac{2TP}{2TP + FP + FN}$$

Precision, which is the ability of an algorithm to identify only relevant instances, has a value of 99.98% and 99.74%. In contrast, recall, which measures the extent to which the algorithm correctly identifies cases, has the same value as the F1 measure, which is the harmonic mean of precision and recall, and also produces a value of 100% and 99.77%. This means that there is a high level of precision and recall, thus suggesting that the model minimises false positives and negatives (see Tables 4 and 5).

The fact that there are minimal false positives or negatives, even with a deficient percentage as shown in Figs. 4, 5, 12, 13, proves the model's viability in natural IoT systems. What is optimal in this case is an abrupt anomaly detector that can capture all anomalies without producing false negatives or false positives, thus maintaining the security of IoT networks.

Figs. 14 and 15 demonstrate the training and validation accuracy graphs for the FCNNs model proposed in the ToN-IoT dataset and the UNSW-NB15 datasets. Note that the x-axis has epochs as its label and the y-axis has accuracy. The training and validation accuracy for the ToN-IoT dataset started from 0.5 and increased significantly to about 0.9998 in the 10th epoch. Subsequently, the validation and training accuracies for the UNSW-NB15 dataset start from 0.5, increase gradually, and reach the value of 0.9974

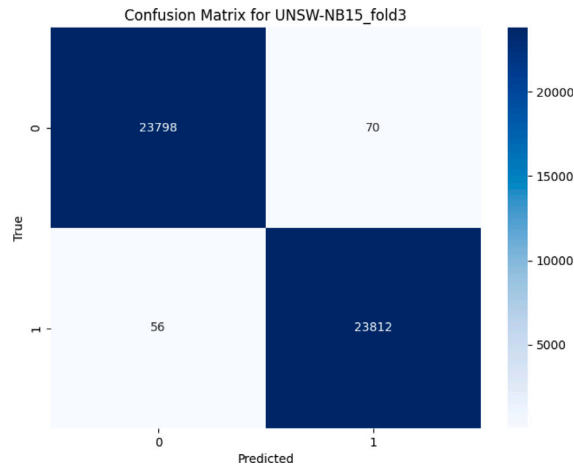


Fig. 13. FCNNs Confusion matrix of the FCNNs model on the UNSW-NB15 dataset: The matrix portrays the classification results for the UNSW-NB15 dataset using the proposed FCNNs model. The model achieved 23,798 true positives, 23,812 true negatives, 70 false positives and 56 false negatives, exhibiting excellent precision and recall.

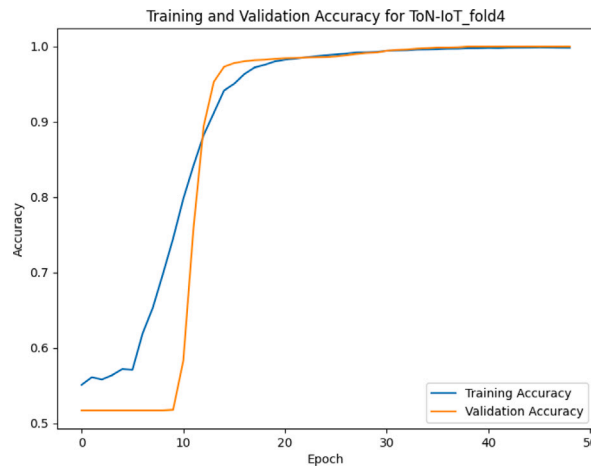


Fig. 14. FCNNs ToN-IoT Training and Test Model accuracy over training epochs.: Accuracy curve of the FCNNs model on the ToN-IoT dataset. As soon as the model switched to the second epoch, the accuracy converged very quickly, and both training and validation accuracies reached 99.98% and stabilised by the 10th epoch. The curves being very close indicates a good amount of generalisation without overfitting.

at the 100th epoch mark. The training and validation accuracy curves being close to each other indicates that the model neither overfits nor underfits the unseen data. The accuracy achieved in this work is 99.98% for ToN-IoT and 99.74% for UNSW-NB15 with a reasonably low effort, indicating that the proposed FCNNs architecture is effective and efficient for anomaly detection in various IoT environments.

Figs. 16 and 17 present the loss value curves for training and validation of the proposed FCNNs model on the ToN-IoT and UNSW-NB15 datasets. On the x -axis, one can find epochs, while on the y -axis, the value of loss is displayed. Starting with the ToN-IoT dataset, the training and validation loss values began at 0.7 and steadily converged to 0.1 by the 50th epoch. Similarly, for the UNSW-NB15 dataset, the loss values began at around 0.8 markers, and by the 100th epoch, they converged to 0.1. The results of both datasets suggest that the model is trained without overfitting, as evidenced by the continuous decline of training and validation losses (see Figs. 18 and 19).

4.2. Performance analysis using ROC and learning curves

In addition to conventional metrics, Figs. 20 and 21 provide a general visual evaluation of the suggested FCNNs on both datasets. In ToN-IoT (Fig. 20a–c), the baseline XGBoost ROC curve demonstrates perfect separability ($AUC = 1.0$), and our FCNNs performs nearly the same in validation and test sets, confirming its ability to generalise beyond training data. The ROC-AUC learning curve (Fig. 20d) exhibits rapid convergence to the near-unity AUC within the first 30 epochs, highlighting the efficiency of the training

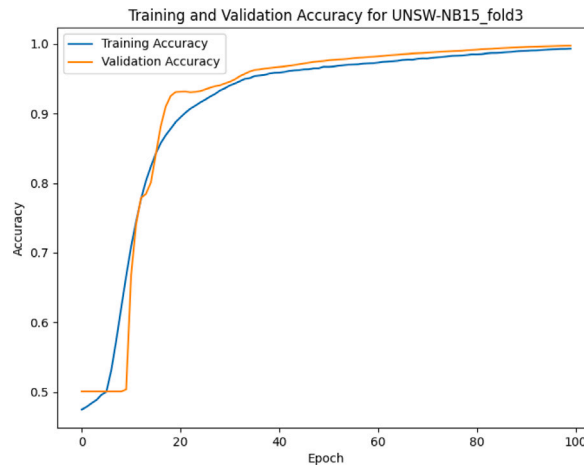


Fig. 15. FCNNs UNSW-NB15 Training and Test Model accuracy over training epochs: UNSW-NB15 DataSet using the trained FCNNs model with accuracy curves of training and validation. This plot shows a steady convergence of accuracy, with the accuracy reaching over 99.74% around the 100th epoch for both training and validation. It also shows that the model heuristically generalises well and does not overfit.

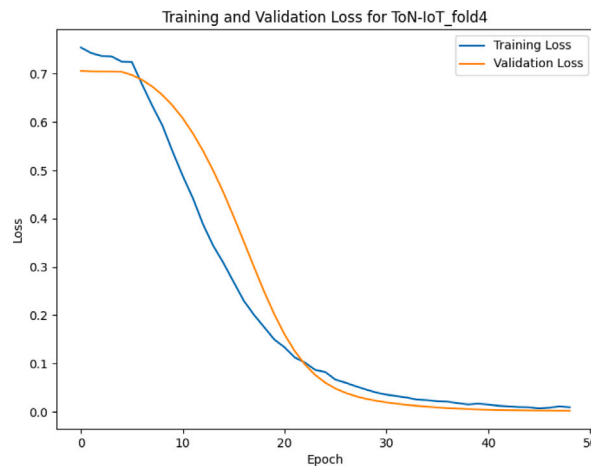


Fig. 16. FCNNs ToN-IoT Training and Test Model loss over training epochs: Training and validation loss curves of the proposed FCNNs model on the ToN-IoT dataset. The plot indicates a gradual decline in loss, confirming that the training loss and validation loss converge to around 0.1 by the end of the 50th epoch.

model. In UNSW-NB15 (Fig. 21a–c), FCNNs achieves substantial gains over the baseline, with an AUC of 0.9346 in the validation set and 0.9857 in the test set, a remarkable result given the inherent class imbalance of the dataset. The learning curve (Fig. 21d) also illustrates stable optimisation and good discriminative power during training. These results collectively provide strong evidence that the proposed FCNNs can achieve high-fidelity anomaly detection with computational efficiency, making it highly desirable for real-time IoT security monitoring.

4.3. Model interpretability using SHAP

To clarify the results, Figs. 1 and 2 show SHAP summary visualisations (Shapley Additive Explanations) for the ToN-IoT and UNSW-NB15 datasets, respectively, providing interpretable explanations for the internal decision-making process of the proposed FCNN model. In Fig. 1, the model relies on attributes such as attack_cat, id, dttl, and sbytes, while the colour gradient represents feature values and their directional effect on anomaly detection labels. Large values for vital indicators, such as attack_cat and id, significantly contribute to classification boundaries, verifying that the model effectively captures malicious signatures. In sharp contrast, Fig. 2 highlights the system performance counters, including the Process_Virtual_Bytes Peak and various LogicalDisk(Total) quantities, which significantly influence model predictions. At a hardware level, these so far underexplored signals capture subtle variations in the system behaviour characteristic of advanced attacks. The clear separation and magnitude of SHAP quantities in both figures not only verify feature selection but also highlight the model's interpretability, transparency, and robustness

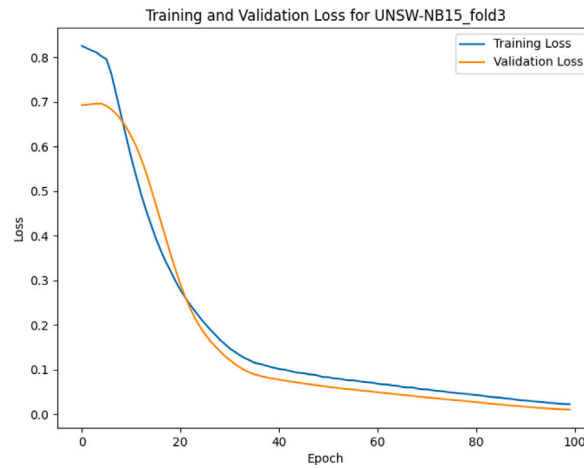


Fig. 17. FCNNs UNSW-NB15 Training and Test Model loss over training epochs: Training and validation loss curves for the proposed FCNNs model on the UNSW-NB15 dataset. The plot also shows a gradual decline in the loss, reaching about 0.1 for both training and validation losses around the 100th epoch. The continued drop indicates that learning is occurring without overfitting.



Fig. 18. SHAP summary plot for the ToN-IoT dataset illustrating the impact of the top 20 features on the FCNN model's output in detecting IoT-based cyber threats.

concerning varying datasets. These explanations further reinforce the generalisability of the model, verifying its readiness for practical deployment in various IoT security applications.

4.4. Inference time analysis

To evaluate the real-time feasibility of the proposed FCNN model, we measured its average inference time per sample in the destination hardware setup, which consisted of an NVIDIA Tesla T4 GPU with 32 GB of RAM in a CUDA-enabled environment. For

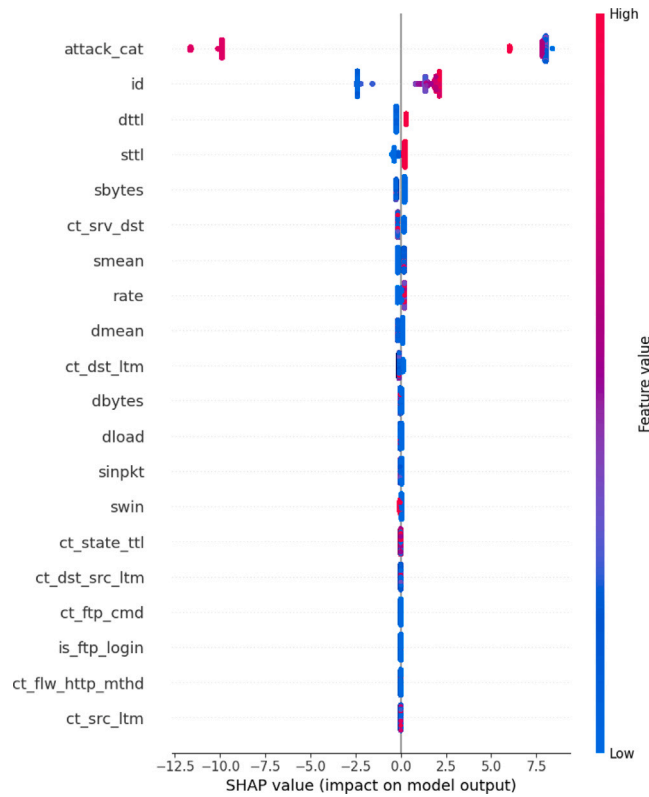


Fig. 19. SHAP summary plot for the UNSW-NB15 dataset revealing the importance of key system-level features influencing the FCNN model's anomaly classification performance.

Table 6

Average inference time per sample for the proposed FCNN model on the target GPU hardware, demonstrating sub-millisecond latency, confirming suitability for real-time IoT anomaly detection.

Dataset	Avg. Inference time per sample (ms)
ToN-IoT	0.0572
UNSW-NB15	0.0865

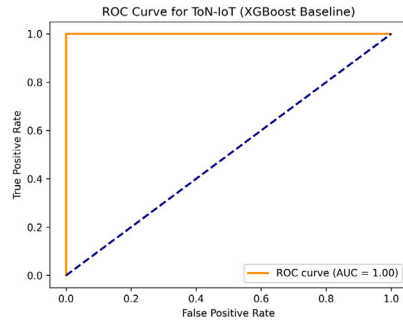
every dataset, five independent forward-pass runs were performed, and a single warm-up run was performed to prevent any latency due to system initialisation.

As shown in Table 6, the results show that the model incurs a very low inference latency of under 0.1 ms per sample for both datasets. This translates to a throughput of more than 11,000 predictions per second, showing the computational efficiency of the model. This makes the model highly suitable for real-time IoT security use cases, where the identification of malicious activity must be accurate and rapid. The model's speed rests primarily in its use of lightweight residual blocks, attention mechanisms, and optimised inference pathways.

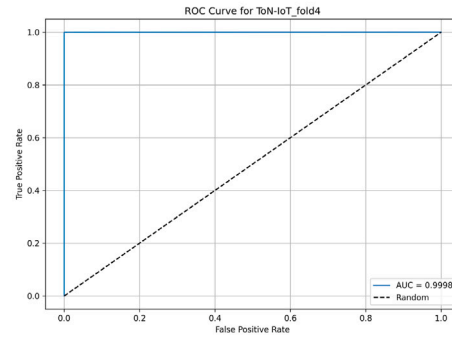
4.5. Learning dynamics

The training, test, and loss curves indicate that the model performs well and converges effectively, as shown in Figs. 14–16, and 17. The training accuracy plots show that the model accuracy in the training set begins to increase from nearly zero per cent and quickly stabilises at 99.98% and 99.74%, respectively. The test accuracy plots show a similar trend, but slightly higher. The loss plot also indicates a general reduction in training and test loss over epochs, which means that the model learns the proper patterns without 'memorising' the data.

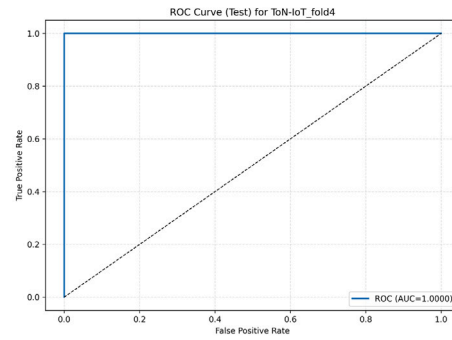
From these performances, it can be concluded that the FCNNs model has indeed generalised well on the test data and performs well on different data sets. The pivotal advantage of this generalisation capability is its applicability in real-world scenarios where the model will encounter unseen data.



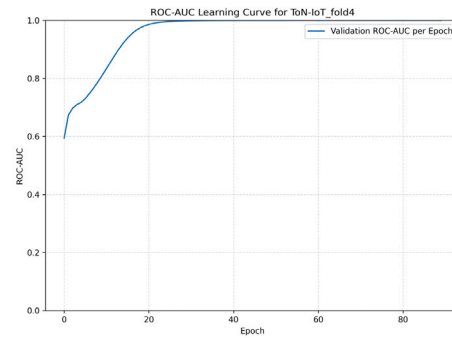
(a) Baseline ROC (XGBoost).



(b) FCNN ROC (validation set).

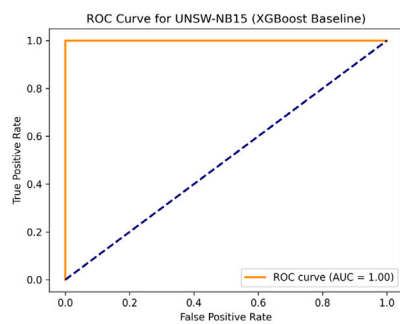


(c) FCNN ROC (test set).

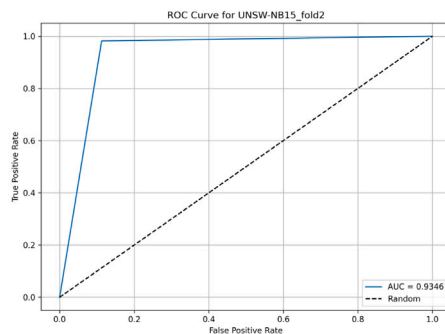


(d) ROC-AUC learning curve (per epoch).

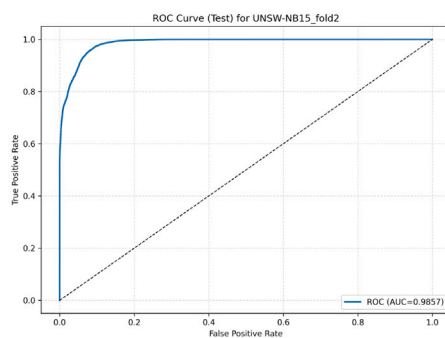
Fig. 20. ToN-IoT: ROC analyses and learning dynamics. (a) Baseline XGBoost ROC, (b) FCNN validation ROC, (c) FCNN test ROC showing generalisation, and (d) validation ROC-AUC versus epoch, indicating rapid convergence.



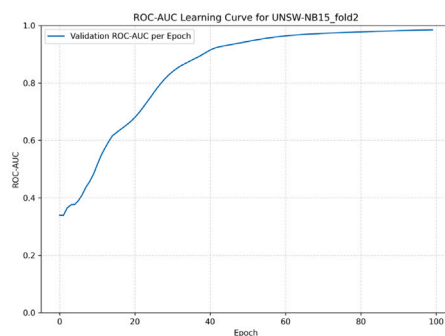
(a) Baseline ROC (XGBoost).



(b) FCNN ROC (validation set).



(c) FCNN ROC (test set).



(d) ROC-AUC learning curve (per epoch).

Fig. 21. UNSW-NB15: ROC analyses and learning dynamics. (a) Baseline XGBoost ROC, (b) FCNN validation ROC, (c) FCNN test ROC demonstrating improved separability, and (d) validation ROC-AUC progression indicating stable convergence.

5. Discussion

Recent research has confirmed the importance of testing deep learning-based anomaly detection models against simpler baseline methods to ensure resilience, interpretability, and efficiency in IoT systems. Ranpara et al. [52] developed a hybrid deep learning system that incorporates semantic reasoning to respond to threats in real time. Their solution was approximately 98.5% accurate, but was more complex in terms of architecture and did not allow for modular interpretation. A.R and Katiravan [53] also proposed a deep neural network model that works with the blockchain to make anomaly detection more transparent. However, it needed more computing power and was not as accurate as our technique. Rhachi et al. [54] used deep autoencoders with ANOVA-based feature selection to achieve a detection rate of around 92.5%. This approach was more effective than typical machine learning methods, but still fell short of high-precision standards. Recent studies have employed federated learning in conjunction with blockchain-inspired solutions to achieve robust and privacy-enhanced one-class learning in IoT networks [55–57]. Although they show promising results, they often introduce complexity in implementation due to the use of distributed computation configurations. Compared to BFKD [55], which employs federated distillation using blockchain-based methods, our method eliminates the overhead of cross-device communication while achieving a precision of 99.97%. Similarly, while APDPFL [56] discusses the incorporation of differential privacy in a decentralised manner, our centralised system uses SHAP feature selection for greater interpretability without a decrease in performance.

In contrast, our research presents an enhanced FCNN architecture that surpasses others due to SHAP-guided feature selection, multi-head attention, and residual connections. The model surpasses all previous findings in similar situations, achieving 99.98% and 99.74% accuracy in the ToN-IoT and UNSW-N15 datasets, respectively. ROC curves (Figs. 20 and 21) demonstrate that the proposed FCNN achieves near-optimal separability for ToN-IoT and offers a reasonable improvement over the baseline for UNSW-NB15. ROC-AUC learning curves show discriminative ability within the first 30 epochs, indicating training efficiency and readiness for real-time deployment. Our approach also addresses real-world deployment and performance challenges highlighted in recent research by maintaining computational efficiency, promoting model explainability, and incorporating rigorous testing against traditional models.

5.1. Comparison with previous work

The results of this model show significant improvements in several key areas compared to previous studies in the field.

5.1.1. Accuracy and precision

The proposed FCNN model achieves an impressive 99.95% accuracy rate in the ToN-IoT dataset and 99.74% in the UNSW-NB15 dataset, demonstrating its strength in various IoT settings. Its speed and accuracy make it efficient for IoT devices with limited resources. This precision is higher than that presented in many earlier studies. For example, Dai et al. [43] obtained results at a level of 98.7% in terms of precision using a deep learning-based model for detecting IoT anomalies, and Ahmad et al. [9] achieved an accuracy of 97.8% in a study conducted in 2020 of a task similar to the deep neural network (DNN). This enhanced performance has been achieved due to the advanced configuration of the current model, which easily maps the intricate dependency between the features in IoT data.

5.1.2. Balanced performance

The confusion matrix indicates that there is almost perfect separation, with only six false positives, while there are no instances of false negatives. This balance is critical in cybersecurity applications because losing anomalies (false negatives) is significantly more expensive than false alarms. The current FCNNs model can contribute to a better detection mechanism than that of Mishra et al. [34], which used DCGAN with an accuracy of 97.3%; however, this poses a high probability that real-world threats will be missed.

5.1.3. Scalability and computational efficiency

Both the training and test accuracy plots present figures not only of the model's performance but also its scalability. The model is not overfitted, exhibiting consistent performance across different epochs, in contrast to other models, such as those of Reddy et al. [11], which used deep neural networks for urban IoT. However, such models tend to have poor scalability when applied to a large dataset or even more complex IoT contexts. However, FCNNs remain stable and can thus be used in real-time applications across various IoT scenarios.

5.1.4. Loss convergence

The loss plot shown in Fig. 16 confirms the decrease in training and test loss without overfitting the model to the input data. This is particularly significant compared to conventional paradigms or enhanced deep learning approaches, such as those of Protogerou et al. [45], who used graph neural networks (GNNs). Although GNNs can learn graph-structured data, their training is computationally expensive and convergence is not always as smooth as the presented FCNNs.

5.2. Limitations and challenges

Despite the high performance of the proposed FCNNs method, we need to investigate some limitations. The existing design may be susceptible to adversarial examples and might not adapt to new attack traces. Furthermore, although residual blocks and dropout enhance generalisation, overfitting is still possible, especially with smaller or imbalanced datasets. In our future work, we will explore the combination of adversarial training and continual learning methods to improve the robustness and adaptability of heterogeneous IoT networks.

5.3. Implications for IoT cybersecurity

This study can be used to draw connections and possible implications for IoT security policies. The accuracy and F1 scores demonstrate that FCNNs can effectively locate anomalous instances, and the results are almost comparable across the different IoT settings. Such a placement puts FCNNs in a stable position for security enhancement in IoT networks, specifically for resource-limited devices. Furthermore, robustness in terms of accuracy and loss over iterations, resulting in different data distributions, implies that the model can be easily implemented in real-time within an IoT environment to perform anomaly detection tasks in existing infrastructure. This is crucial to ensure the safety and security of IoT systems, particularly in organisations and industries that require the utmost security and safety of their data and systems, including healthcare, industrial IoT, and smart cities.

5.4. Class imbalance

To address the imbalance in the class, a common issue in anomaly detection datasets, we oversampled the minority class using the Synthetic Minority Oversampling Technique (SMOTE) in the initial stage. SMOTE creates new samples for the minority class by blending existing instances, ensuring that both the majority and minority classes are represented proportionally. This strategy is beneficial for detecting IoT anomalies, where anomalous behaviour is often outnumbered by normal behaviour. The impact of SMOTE is quite clear from the precision and recall of both notably high classes. For example, in the ToN-IoT dataset, the model achieved 1,998 true positives, 2,221 true negatives and only two false positives, with no false negatives. This signifies excellent performance in the face of class imbalance, as the model can perform well even in the presence of this imbalance. Likewise, the same model working on UNSW-NB15 had 23,798 true positives, 23,812 true negatives, 70 false positives, and 56 false negatives. These results further demonstrate the model's ability to operate with imbalanced data. Such promising results highlight the critical nature of the class imbalance problem, which, if left unaddressed, could lead to negligence in addressing the class, which is of utmost importance for accurate anomaly detection.

5.5. Adversarial examples

Although adversarial examples were not part of the prospective tests in this investigation, they pose a significant problem for anomaly detection systems in an IoT environment. Adversarial examples are presented as inputs that intend to confuse machine learning models by hiding slight deviations within those inputs. Attacks such as these threaten systems designed to detect changes, as they can result in reduced accuracy or complete failure to detect a specific anomaly, instead leading to the detection of a false one. An example is an attacker who creates adversarial examples to help bypass security systems or trigger false alerts. Defeating the system attempts to circumvent detection and violate its operating principles. To make the model more robust against adversarial attacks, future research could investigate the use of adversarial training methodologies, such as the fast gradient sign method or projected gradient descent methods. In these techniques, the model is trained in a combination of benign and adversarial examples to learn features that can withstand adversarial perturbation. The model could also be improved by using defence distillation or obfuscation to achieve more powerful adversarial-resistant models. Dealing with adversarial examples is a critical issue to consider in the context of security and trust in the integrity of IoT systems, and anomaly detection systems in particular, especially in the most practical environments where attackers seek to exploit model flaws.

6. Future research direction

However, several critical research gaps still need to be addressed in the context of anomaly detection models for IoT networks, which hinder the development of current solutions. Among such gaps, one of the most severe deficits is model interpretability. Although approaches like Fully Connected Neural Networks (FCNNs) are accurate because they are almost at the human level, achieving an accuracy rate of 99.8%, they lack "interoperability" or, as we refer to it, interpretability, that is, no one can explain how this model arrives at the decision it has made, even though in cases of cybersecurity, trust, and regulatory compliance outweigh efficiency. Furthermore, most models are trained and validated using data that is in a static state, while in IoT scenarios, data is real-time in nature. The nature of IoT data is also dynamic, resulting in concept drift – a change in data distribution — that renders many static models ineffective. This results in a concept drift, which is a change in the distribution of the data. In addition, many current solutions are highly computational and have not been highly optimised for IoT end devices networks or at edges, for example. Other problems are a lack of models that can withstand adversarial attacks, a lack of robust models in different IoT environments, and scalability issues on large IoT networks.

The proposed FCNNs architecture is developed explicitly for an IoT-enabled environment and addresses the anomaly detection problem. The proposed model achieves high accuracy in anomaly detection, with metrics of 99.98% and 99.74%, respectively. Furthermore, the model is designed with low computational complexity in mind, making it suitable for resource-constrained IoT devices. Furthermore, this study emphasises the preprocessing of IoT data and the methodological techniques involved in managing large and complex datasets. It also provides a starting point for the practical implementation of deep learning in IoT security and motivates the development of lightweight yet effective deep learning anomaly detection systems. The study encourages further research on enhancing neural network architectures and hybrid anomaly detection systems in IoT environments.

6.1. Real-time anomaly detection in dynamic environments

A dependence on static datasets is a drawback of the current study, as IoT environments are dynamic and prone to concept drift. Later studies might consider investigating techniques for real-time anomaly detection as a solution. For example, Gama et al. [58] proposed online learning algorithms which could be modified to suit the FCNN architecture for continuous adaptation to shifting data distributions. Furthermore, streaming data technologies such as Apache Kafka and Apache Flink could be employed to capture and process IoT data in real-time. In this way, the effectiveness of the model in anomaly detection is preserved even as the underlying data distribution changes, enabling more realistic real-world IoT use cases.

6.2. Enhancing model interpretability

The results confirmed this since the FCNNs model scored almost perfect accuracy; nevertheless, the interpretability of the results is a significant issue with such black-box approaches. More research is needed in modelling the DL based architecture of IoT anomaly detection, while the model must be computationally interpretable. More specifically, scholars focusing on FCNNs models should incorporate XAI techniques to increase the interpretability of the decision-making processes of models. This would help stakeholders understand why it is possible to refer to certain events as anomalies, which is essential for establishing trust and ensuring regulatory compliance in cybersecurity.

6.3. Adversarial robustness

An added limitation in the model stems from the susceptibility of the model to adversarial attacks, which is a technique to test the security of machine learning algorithms by generating deceptive inputs. Further research can also investigate the potential of adversarial training using techniques such as the FGSM (Fool's Gold Sign Method) of Goodfellow et al. [59], or PGD (Projected Gradient Descent) of Madry et al. [60], to improve the accuracy of the model in these hostile tasks. Adversarial training refers to the concept of training a model to make it more robust to changes under attack. Moreover, it is observed that, according to Papernot et al. [61], the model can be improved by distillation, as one method to increase its robustness is to force smoother output distributions from the model. In the context of IoT anomaly detection, addressing adversarial robustness is important for providing security and reliability in real-world usage, as malicious users would try to use unprotected aspects of the model to inflict harm.

6.4. Resource-efficient deployment

The accuracy and computational efficiency demonstrated by the proposed FCNNs model are remarkable; however, deploying it to resource-constrained IoT devices is challenging. Future studies may investigate employing model compression techniques, such as pruning [62] or quantisation [63], to reduce the model's dimensions and computational needs while maintaining effectiveness. Pruning is the process of cutting excessive weights or neurones from the model, while quantisation is the process of lowering the accuracy of the model's parameters. These methods can facilitate the application of high-quality abnormality detection models on edge devices with limited processing capabilities and memory, making it more feasible for practical IoT use cases.

6.5. Generalisation across diverse IoT environments

The model performs exceptionally well in the ToN-IoT and UNSW-NB15 datasets; however, its performance in other IoT environments is not validated. To ensure the feasibility of the model on a larger scale, it would be prudent to evaluate smart cities, industrial IoT, or healthcare IoT datasets. Furthermore, transferring the newly acquired knowledge could allow the model to be used in different environments with minimal training sessions. This strategy would further increase the applicability of the model and guarantee its usefulness across extensive IoT environments.

6.6. Integrating privacy-preserving mechanisms

IoT networks process sensitive data; however, the current model lacks methods that protect client privacy. Further studies can be conducted to implement privacy-preserving approaches, such as federated learning, which enables training models across various devices without transmitting raw data. This would enhance privacy while also helping to detect client privacy anomalies, a crucial feature in use cases such as healthcare, finance, and other IoT applications that require high levels of confidentiality.

6.7. Mitigating ethical and regulatory issues

Detection accuracy can remain high, while the proposed model can be integrated with federated learning to enhance data privacy. In simple terms, federated learning refers to the practice of training models on different devices or servers independently, without sharing raw data with a central location. This approach is beneficial when it comes to complying with General Data Protection Regulation (GDPR) because it reduces the chance of a data leak occurring and protects critical information that can only reside on local devices. The model allows for the capture of anomalies in IoT settings using machine learning, while infringing user privacy makes it possible for the model to be used in medicine, finance, and other industries that require strict confidentiality [64]. Furthermore, incorporating privacy-protective features, such as differential privacy, enhances the model's compliance with ethical and legal requirements. Individual data points on users can be protected from being reverse-engineered by adding noise to model updates or data during training through the use of differential privacy. This method is beneficial in IoT environments where personal devices gather data that needs to be protected, as noted in [65]. In addition to privacy concerns, the proposed model must address other ethical issues, including bias and fairness in the detection of anomalies.

Like other AI models designed for real-life applications, IoT models can experience bias arising from unequal sampling, which can lead to unfair smear results. Addressing bias in machine learning systems is always preferred, as system fairness is a necessary approach [66]. Trust decreases as model accuracy increases with transparency; this is also the case with regulatory compliance. Stakeholders are rarely satisfied with the available level of accuracy of the FCNNs model, as its “black box” poses a huge difficulty in understanding the logic behind the decisions made. XAI methods like SHAP (SHapley Additive Explanations) leave no room for debate about how models capture and explain anomalies, which can be verified and invalidated by auditing processes [67]. Informing users and stakeholders about trust is essential within the building regulatory environment.

6.8. Hybrid approaches for enhanced detection

More studies are needed to propose hybrid systems that integrate FCNNs with other types of Artificial Neural Network (ANN), such as RNNs for temporal data or transformers for sequential data. Future work could explore a combined scheme to improve the identification of intricate, composite, and multimodal anomalies, thereby improving the safety and privacy of transmitted data in IoT security.

7. Conclusion

In this research, we propose to improve the structure of FCNNs for detecting anomalous IoT network traffic. The introduced model proved to be highly effective in detecting anomalies based on IoT data, due to its high precision and recall rates, combined with relatively low computational complexity compared to traditional approaches. Specifically, it achieved an accuracy of 99.98% and an F1 score of 99.95% on the ToN-IoT dataset, as well as an accuracy and F1 Score of 99.74% on the UNSW-NB15 dataset. This performance demonstrates that this approach is suitable for real-time applications, such as IoT settings, because of the immediacy of threat detection.

Our study demonstrates that the model is scalable and extensible, overcoming the heterogeneity, compliance, high dimensions, and ever-changing threat changes characteristic of IoT data. The analysis yields encouraging results; nevertheless, some aspects require future research. The near-unity AUC values and convergence exhibited in Figs. 20 and 21 confirm the model's ability to deliver real-time anomaly detection with excellent generalisation, which justifies its deployment in real-world IoT applications. Based on the results of this work, future research should involve providing more extensive information on the model and its real-time implementation to identify emerging anomalies in IoT settings. However, the direction of lightweight model adaptations for deploying the models presented in resource-limited IoT endpoints, incorporating privacy-preserving strategies and frameworks, could expand the use of the proposed models to various IoT contexts.

In conclusion, the concept presented in this work opens the possibility of building more reliable and robust anomaly detection systems, which can be translated into significant security enhancements for the IoT paradigm. This research on the various layers of the proposed FCNN model demonstrates that it offers sufficient flexibility and effective performance for practical use in real-world IoT environments, thereby enhancing IoT security and reliability.

Abbreviations

The following abbreviations are used in this manuscript:

FCNNs Fully connected neural networks

IoT Internet of Things

ML Machine Learning

DL Deep Learning

CNNs Convolutional Neural Networks

XAI explainable artificial intelligence
NLP natural language processing
DoS Denial of Service
IDS intrusion detection systems
DCGAN deep convolutional generative adversarial networks
TON Telemetry Operating system Network Traffic Dataset
OS operating system
CPU central processing unit
IoMT Internet of Medical Thing
IIoT Industrial Internet of Things
ANN Artificial neural network
ECG electrocardiogram
TP True Positives
TN True Negatives
FP False Posives
FN False nagatives
GRU Gated Recurrent Unit
SVDD Support vector data description
SHAP Shapley Additive Explanations
VAE Variational autoencoder
LPWANs Low-Power Wide Area Networks
LSTM Long Short-Term Memory
GAT Graph Attention Network
MI MutualInformation
DNN Deep Neural Network
DS2OS Distributed Smart Space Orchestration System
RNNs recurrent neural networks
ANN Artificial Neural Network
GDPR General Data Protection Regulation

CRedit authorship contribution statement

Muhammad Muhammad Inuwa: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Resul Das:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Project administration.

Ethical approval

Ethical approval was not necessary because this study did not include humans or animals and did not collect data requiring ethical clearance. Therefore, no ethical approval was required.

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Declaration of competing interest

The authors report no known financial interests or personal affiliations which could have affected their research outcomes.

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Data availability

Researchers can find the datasets from this study online at the following locations:

TON-IoT Dataset: Available from the UNSW Canberra website.

UNSW-NB15 Dataset: The UNSW Australian Centre for Cyber Security (ACCS) provides access to this data.

Researchers may request supplementary data and code from the study's corresponding author by making a reasonable appeal.

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